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Light amplification by stimulated emission of radiation (laser) is a coherent and monochromatic source of electromagnetic radiation that can propagate in a straight line with negligible divergence. As a result, laser finds diverse applications ranging from mere mundane to most sophisticated uses either for totally commercial or purely scientific purposes, and from life saving to life threatening causes. High power lasers can produce intense heating and perform various manufacturing operations or material processing. The present contribution provides an overview of the application of high power laser only for material processing in engineering applications, and intentionally excludes the scope of application of laser in metrology, biomedical technology, spectroscopy, etc. The manufacturing processes covered have been broadly divided into four major categories, namely, laser assisted forming, joining, machining and surface engineering. Besides discussing the scope and principle of these processes, each section enumerates a detailed update of literature, scientific issues and technological innovations. At the beginning, a brief introduction to different types of lasers and their general applications, fundamentals of laser–

the state or the physical properties of the active medium. Consequently, there are glass or semiconductor, solid state, liquid and gas lasers. Gas based lasers can be further subdivided into neutral atom lasers, ion lasers, molecular lasers and excimer lasers. The typical commercially available lasers are: solid state or glass laser (Nd:YAG, ruby), semiconductor or diode laser (AlGaAs, GaAsSb and GaAlSb), dye or liquid lasers (solutions of dyes in water/alcohol and other solvents), neutral or atomic gas lasers (He–Ne, Cu or Au vapour), ion lasers (argon (Ar^+) and krypton (Kr^+) ion), molecular gas lasers (CO_2 or CO) and excimer laser (XeCl and KrF). The wavelengths of the currently available lasers cover a wide spectral range from the far infrared to the soft X-ray. Laser comprises three principal components, namely, a gain medium, a device for exciting the gain medium and an optical delivery/feedback system.[10] Additional provisions of cooling the mirrors, guiding the beam and manipulating the target are necessary to facilitate material processing.

Laser–matter interaction for material processing

The process of energy deposition from a pulsed/continuous wave (CW) laser beam into the near surface region of a solid involves electronic excitation and deexcitation within an extremely short period of time.[11] In other words, the laser–matter interaction within the near surface region achieves extreme heating and cooling rates (10^3–10^{10} K s^{-1}), while the total deposited energy (typically 0·1–10 J cm^{-2}) is insufficient to affect, in a significant way, the temperature of the bulk material. This allows the near surface region to be processed under extreme conditions with little effect on the bulk properties.

The initial stage in all laser assisted material processing applications involves the coupling of laser radiation to electrons within the metal. Initially, photon interaction with matter occurs through the excitation of valence and conduction band electrons throughout the wavelength band from infrared (10 μm) to UV (0·2 μm) regions. Absorption of wavelength between 0·2 and 10 μm leads to intraband transition (free electrons only) in metals and interband transition (valence to conduction) in semiconductors. Conversion of the absorbed energy to heat involves: excitation of valence and/or conduction band electrons, excited electron–phonon interaction within a span of 10^{-11}–10^{-12} s, electron–electron or electron–plasma interaction and electron–hole recombination within 10^{-9}–10^{-10} s (Auger process). Since free carrier absorption (by conduction band electrons) is the primary route of energy absorption in metals, beam energy is almost instantaneously transferred to the lattice by electron–phonon interaction. Similarly, transition in semiconductor or polymers having ionic/covalent bonding with energy gap between conduction and valence bands is marginally slower.[11]

Laser processing of engineering materials

Application of laser to material processing can be grouped into two major classes: applications requiring limited energy/power and causing limited microstructural changes only within a small volume/area without change of state and applications requiring substantial amount of energy to induce the change in state and phase transformation in large volume/area. The first category includes semiconductor annealing, polymer curing, scribing/marking of integrated circuit substrates, etc. The second type of application encompasses cutting, welding, surface hardening, alloying and cladding. The average power or energy input is relatively low in the first category, while that for the second category is higher as the processes involve single or multiple phase changes within a very short time. Almost all varieties of lasers can perform both types of operations in CW and pulsed mode provided appropriate power/energy density and interaction time for the given wavelength are applied.

The domain for different laser material processing techniques as a function of laser power and interaction time is illustrated in Fig. 1.[2] The processes are divided into three major classes, namely, heating (without melting/vapourising), melting (no vapourising) and vapourising. It is evident that transformation hardening, bending and magnetic domain control which rely on surface heating without surface melting require low power density laser. On the other hand, surface melting, glazing, cladding, welding and cutting that involve melting require high power density laser. Similarly, cutting, drilling and similar machining operations remove material as vapour hence, need delivery of a substantially high power density within a very short interaction/pulse time. Since all laser material processing

Table 1 Commercially available lasers and their industrial applications

Laser	Discovery	Commercialisation	Wavelength/nm	Application
Ruby	1960	1963	628	Metrology, medical applications, inorganic material processing
Nd:glass	1961	1968	1060	Length and velocity measurement
Diode GaAs/GaAlAs	1962	1965	780–905	Semiconductor processing, biomedical applications, welding
He–Ne	1962		1152	Light pointers, length/velocity measurement, alignment devices
Carbon dioxide	1964	1966	10 200	Material processing – cutting/joining, atomic fusion
Nd:YAG	1964	1966	1064	Material processing, joining, analytical technique
Argon ion	1964	1966	480–515	Powerful light, medical applications
Dye (sodium fluorescein)	1966	1969	535–600	Pollution detection, isotope separation
Copper	1966	1989	511	Isotope separation
Excimer	1975	1976	300–350	Medical application, material processing, colouring
Free electron laser	1971	1997	2000–10 000	Medical surgery, surface modification of polymer

operations can be defined by an appropriate combination of power density and interaction time, one is tempted to combine these two into a single scalar parameter as energy density (power density multiplied by time, J mm^{-2}) for the sake of simplification and convenience. However, the exercise is bound to prove futile and not advisable as both the quantum of energy and its temporal and spatial interaction with matter (rather than their product) are crucial to achieve the desired microstructural/phase/state changes and properties for a given material. For instance, application of 10^{-2} J mm^{-2} energy density may induce either surface hardening (say, for a given material combination of 10^{2} W mm^{-2} power density and 10^{-4} s interaction time) or surface melting (say, for a combination of 10^{4} W mm^{-2} power density and 10^{-6} s interaction time).

Beam configuration or beam profile also plays an important factor in determining the energy distribution at the interaction zone during laser processing. Four types of beam profiles, namely, Gaussian, multimode, square (or rectangular) and top hat are commonly used for material processing.[2] A Gaussian beam is most suitable for cutting and welding applications rather than for surface treatment because, being a 'sharp tool,' it tends to vapourise and melt the substrate deeply. In contrast, multimode, top hat and square profiles ('blunt tools') are preferred for surface engineering. Beam shaping is the process of altering an input beam to produce an output beam with a desired spatial or irradiance profile. Beam shaping plays an important role in determining the quality of hardening, welding and laser microprocessing.[12,13] The beam shaping techniques may broadly be classified into three categories.[13] The first, and most simple, class of beam shaping is aperturing, which is achieved by using an aperture to allow some portion of the beam power to pass through while the rest of the beam is either reflected or absorbed. A considerable amount of power loss is the major drawback associated with aperturing. In the second type, a field mapper is used to transform an input beam into a desired output beam in a controlled and efficient manner.[13] This technique could be used to transform a

single mode Gaussian beam into a wi38irrTdTTdr. c6(be)-991.2(of)-4 geometry, properties and composition with high precision/productivity are the advantages associated with laser assisted bending. Bending angle and properties of the bent zone may be controlled by controlling parameters such as laser beam radious, power density, interaction/pulse time, material properties (thermal, physical or chemical) and dimension/geometry of the workpiece (thickness, curvature, etc.). Bending of strong and difficult to bend metals (body centred cubic, body centred cubic or hexagonal close packed and hexagonal close packed), intermetallics, composites and ceramics has been an important motivation for the increasing interest in laser assisted bending. The success of laser bending of semiconductor and polymeric sheets is of great interest to the semiconductor and packaging industry. There are three mechanisms of laser assisted bending, i.e. temperature gradient mechanism, buckling mechanism and upsetting mechanism.[25–29] Many applications involve a complex combination of these mechanisms rather than only one of them.

Namba[23,24] was one of the first few to attempt and report the high degree of flexibility of laser assisted bending of AISI 1045 and AISI 304 stainless steel sheets by using a defocused CO_2 laser. Subsequently, several studies on laser bending of steel and other metallic materials have been conducted.[25–28,30] Vollertsen and Geiger[31] described the possibility of laser assisted straightening of welded component in a car body to reduce distortion arising out of welding by numerical modelling and subsequent experimental study. Hennige *et al.*[32] developed a hardware for closed loop controlled laser bending that could produce an accuracy of 0·2° for 10° bending angle, which corresponds to the range of scattering of one single bending operation. Depending on the number of necessary steps to achieve a given bending angle, different types of control strategies could be performed. Geiger[33] described hybrid forming, a combination of conventional forming followed by net shaping by laser forming to combine the speed of conventional forming with the accuracy of laser forming without any special set-up. However, a detailed process planning is required for the large scale application of the hybrid process. Magee and de Vin[34] demonstrated the possibility of accurate bending of mild steel sheet with 0·8 mm thickness using the hybrid technique of bending and subsequently, cutting using the same set-up.

Laser assisted deep drawing is a non-contact process in which a laser beam is used to heat the wire to a critical temperature in order to make the wire plastic and easily deformable. High power laser has been successfully employed for drawing of several metals and alloys. Under optimum processing conditions, nickel wire was found to exhibit an increased drawability as compared with conventional annealing.[35] Drawing limit (in terms of area reduction ratio, i.e. final/initial cross-sectional area) was found to increase to a maximum value and then decrease with increasing incident laser power.[35] This behaviour is attributed to the limit of maximum drawability and plasticity of nickel at a given elevated temperature range. Furthermore, brittle materials can be successfully drawn without any rupture or microcrack using laser. Schuöcker[36] developed a mathematical model to predict the temperature and stress distribution during laser assisted deep drawing, showing a considerable reduction in drawing force.

Deformation behaviour of sheet metal (Ti–6Al–4V) by curved laser irradiated path was studied by Chen *et al.*[37] using finite element method. Temperature, displacement, stress and strain fields were determined using this model followed by analysis of deformation behaviour using thermomechanical principle. The simulated results showed that three-dimensional deformation could occur only on one side of a scanning path where the rigid constraint was relatively lower. On this side, the sheet is slightly extended and thinned down. On the other side of the scanning path, the sheet only produced slight rigid rotation. This is very different from the linear laser bending, which is characteristic of symmetrical bending related deformation on either side. Wu and Ji[38] studied the deformation stress field during the laser bending of sheet metal using three-dimensional thermo–elastoplastic finite element analysis based on the reduced integration method. The stages of laser bending and predicted displacement fields agreed well with the experimental data. Marya and Edwards[39] analysed the laser bending of two titanium alloy sheets, Ti–6Al–2Sn–4Zr–2Mo (near alpha alloy) and Ti–15V–3Al–3Cr–3Sn (beta alloy) using a conduction model with a travelling Gaussian heat source. Temperature and bending angle were predicted and correlated with process parameters. Bending was found to initiate at 0·48 of the melting temperature and attained a maximum value at $\sim$0·65 of the melting temperature. Bending behaviour of Ti using a CW CO_2 laser was investigated by Okuda *et al.*[40] who found that bending angle would primarily depend on laser power and spot diameter. The bending direction tended to change from the laser irradiation side to its opposite side when large laser spot diameter was applied.[40]

Cheng and Lin[41] developed an analytical model which showed an increase in angle of bending with increasing laser power, decreasing scan speed and decreasing sheet thickness. Kovacevic *et al.*[42] developed a three-dimensional finite element simulation including a non-linear transient and indirectly coupled thermal–structural analysis accounting for the temperature dependence of the thermal and mechanical properties of the materials. The bending angle, stress–strain relation, temperature and residual stress distribution were predicted by the simulation for different materials, sheet thicknesses, scanning speeds and laser powers. The experimental results were suitably compared with the theoretical prediction. A selected number of important studies on the mechanism of laser bending, process description and theoretical modelling have been reported elsewhere.[28,43–54]

Laser assisted bending of ferrous metals and alloys

Iron and steel happen to be the earliest and most widely explored material to study the usefulness and mechanism of laser assisted bending.[55–74] Effects of the plastic anisotropy in cold rolled mild steel on bending deformation during the laser forming process have been investigated experimentally and numerically by Peng and Lawrence.[55] The variables considered were laser power, scanning speed and number of scans for sheets with different degrees of reduction by prior cold rolling. The simulation results are consistent with the experimental observations.[55] Wu and Zheng[56] studied the effect of process parameters on bending angle in laser assisted bending of AISI 1008 steel. Bending angle was found to be directly proportional to the laser power and number of repetition and inversely proportional to the scan speed and beam size/diameter.[56] Yau *et al.*[57] used a low power Nd:YAG laser to study the bending behaviour of a thin alloy steel strip used for integrated circuit lead frames. Bending occurred at the threshold line energy of 0·14 J mm^{-1}, beyond which a linear relationship between bending angle and laser energy was observed and attributed to the requirement of a critical temperature for thermal upsetting or permanent change of shape.[58] Lawrence *et al.*[59] studied bending of mild steel sheets using a 2·5 kW high power diode laser. The bending angle varied from 0·5 to 40° and increased with increasing applied power, increasing number of passes and decreasing scan speed. It was found that minimum threshold energy was necessary to initiate bending. However, the bending angle did not show appreciable change at very high input energy. Yoshioka *et al.*[60] studied the bending behaviour of AISI 304 austenitic stainless steel wire (0·1 mm diameter) to develop complex frame structures such as zigzag or crank wire frames with an Nd:YAG laser offering 0·02–0·2 J energy. Dutta Majumdar *et al.*[61] carried out a detailed investigation on laser assisted bending of AISI 304 stainless steel sheet of different thicknesses by CW CO_2 laser to determine the extent and rate of bending as a function of laser parameters. The rate of bending increased with increasing applied power density attributed to increased material flow at a higher incident energy and hence, increased thermal distortion associated with the process. However, an optimum range of laser power should be selected so that the applied power density is capable of bending without excess melting, evaporation or crater formation. Wang *et al.*[62] studied the thermophysical process of bending of St C45 steel (C: 0·08, Si: 0·3–0·1, Mn: <0·4, P: <0·025 and S: <0·025) by a CW CO_2 laser. The dynamic microdeformation of the specimen was measured using a laser beam reflex amplifier system. The final deformation angle was found to depend on a combined action of thermal strain and phase transformation strain.

Tube bending is important in manufacturing of boilers, engines, heat exchangers, air conditioner, tube and pipe products. Li and Yao[63] established the mechanism of laser bending of tube. It was found that bending is primarily achieved through thickening of the scanned region instead of thinning of the unscanned region, and the scanned region assumes a slightly protruded shape. The absence of appreciable wall thinning is one of the major advantages of laser bent tubes. The bending radius increased with the beam diameter and the ratio of the tube outer diameter to the wall thickness. Furthermore, the bending efficiency increased with the maximum scanning angle up to a critical value. Hao and Li[64] developed a numerical model to establish a relationship between the bending angle and processing parameters.

Laser assisted bending of non-ferrous metals and alloys

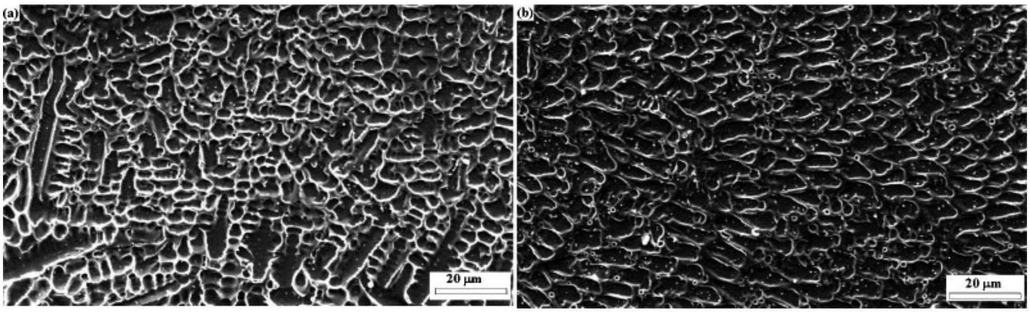
Liu *et al.*[75] studied the deformation behaviour of the SiC reinforced 6092 aluminium matrix composite using a 2 kW Nd:YAG laser. The bending angle was found to decrease with increasing scan speed because of the decrease in interaction time between the laser beam and composite. On the other hand, the bending angle increases with increasing applied power. Under a suitable condition for bending, a linear relationship between the bending angle and the number of irradiation passes was observed. Ramos *et al.*[76] investigated the microstructures of Alclad 2024-T3 Al–Cu alloy following bending by a CO_2 laser. It was observed that the irradiated zone experienced different stages of thermal annealing including recovery (subgrain formation), recrystallisation and grain growth depending on the deposited thermal energy. Recrystallised zone suffered partial melting at the grain boundary corners or triple points leading to precipitation along grain boundaries on cooling. Yau *et al.*[77] carried out laser bending of Al based metal matrix composite (A2009/SiC) by a high power (2 kW) CW Nd:YAG laser and determined the optimum processing conditions for attaining the maximum bending angle. Similar studies on deformation behaviour of laser assisted bending of aluminium based metal matrix composite (Al 6013/SiC_p) sheets showed that bending could be achieved only under a narrow processing range and increased with increasing laser power, decreasing processing velocity and increasing number of irradiations.[78] In another study, it was observed that bending angle was inversely proportional to the volume fraction of SiC.[79] Magee *et al.*[80] studied the bending behaviour of AlCuMg alloy and observed that bending rate/angle primarily depended on temperature and decreased with the number of pulses due to material accumulation at the bend.

Laser assisted bending is a unique technique to bend the difficult to deform metals and alloys. Okuda *et al.*[81] studied the deformation mechanism of Mg alloy by finite element modelling. Chan *et al.*[82] compared the deformation behaviour of chromium sheet having limited room temperature ductility subjected to roll compression and laser bending technique. In roll compression test, a bending angle of 90° was achieved without fracture at a temperature of 100°C. However, the maximum bending angle achieved by non-contact laser bending was 23·5° and the same increased with increasing applied power density and number of irradiation. In spite of its poor room temperature ductility, Ti_3Al based intermetallic alloy was reported to be laser bent by Chan and Liang[83] primarily due to the differential degree of thermal expansion of the intermetallic across the thickness.

Laser assisted bending has been studied in Ti and its alloys.[83–90] Magee *et al.*[84–86] studied the influence of process variables on laser bending of $\alpha+\beta$ Ti alloy (Ti–6Al–4V) using a CW CO_2 laser. Blake *et al.*[87,88] studied the effect of beam divergence, feedrate, laser power and pulse width on the angle and range in laser assisted bending of Ti–6Al–4V and Ti–15V–3Cr–3Sn–3Al using a 400 W pulsed Nd:YAG and a 50 W desktop CO_2 laser. Similarly, Maher *et al.*[89] and Hatayama and Osawa[90] investigated the effect of process variables on the bending behaviour of pure Ti. Marya and Edwards[91] studied the interactions between process variables, materials properties and residual angular distortions (bending angle) during laser assisted bending of Ti–6Al–2Sn–4Zr–2Mo and Ti–15V–3Al–3Cr–3Sn alloys.

Laser rapid prototyping

One of the most recent applications of laser in material processing is the development of small, complex and intricate components by coupling laser with computer controlled positioning stages and computer aided engineering design.[2,92–96] This technology is based on



2 Scanning electron micrographs of top surface of laser fabricated AISI 316L stainless steel with power densities of *a* 0·031 kW mm^{-2} and *b* 0·091 W mm^{-2} and at scan speed of 2·5 mm s^{-1}, and powder feedrate of 136 mg s^{-1} (Ref. 115)

repetitive deposition and processing of material layers known as additive freeform fabrication technique.[97–100] The advantages offered by these techniques include the capabilities to produce components of intricate shape with a greater accuracy, faster processing speed, economy in energy and material consumption, etc.

Laser engineered net shaping is a process in which near net shape metal structures are built from powdered metal layer by layer from computer generated designs. Zhang *et al.*[101] discussed the characteristics of laser engineered net shaping, directed light fabrication and shape deposition manufacturing. The fundamental of selective laser sintering has been studied by Goode.[102] Dai and Shaw[103] developed a three-dimensional thermomechanical finite element model to investigate the transient temperature and residual stress fields in laser assisted fabricated part. The model system chosen was 93Cu+7Sn and NiMo mixed with 70%(MoW)2NiB2:30%NiMo. McAlea *et al.*[104] discussed the laser based rapid prototyping technique and compared with the sand and investment casting technique. The possibilities of application of laser assisted deposition in fabrication of part in packaging industry and microelectronic component have been discussed by Bauer *et al.*[105] and Micheli and Boyd[106] respectively. Griffith *et al.*[107] described the use of contact and imaging techniques to monitor the thermal signature during laser engineered net shaping. The understanding of solidification behaviour, residual stress and microstructural evolution with respect to thermal behaviour was discussed.

Metal rapid prototypes and fabrication of moulds using hybrid processes of selective laser cladding and milling techniques have been studied by Jeng and Lin.[108] The selective cladding process was employed to build up the material layer by layer, and the milling operation was employed to level the top surface of the clad to increase the accumulation of powder and accurately control the clad height. Similarly, the fabrications of three-dimensional structure using microelectrodischarge machining and laser assembly and application of laser welding of wire, subsequent deposition and milling for rapid prototyping were discussed by Kuo *et al.*[109] and Choi *et al.*[110] respectively.

Laser assisted rapid manufacturing of ferrous components

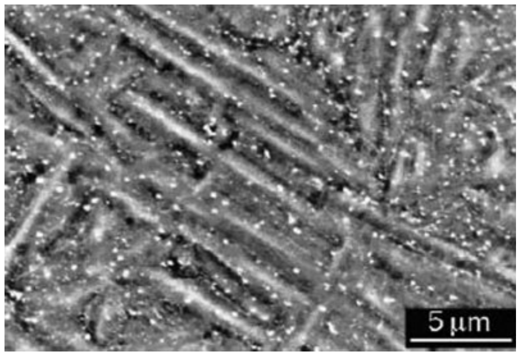
Laser assisted rapid manufacturing has been extensively studied in iron and steel.[108–120] Direct laser sintering of iron, Cr–V tool steel, 7Fe–4·3C, 97Al–3Fe, 93·3Cu–2·8Sn–3·9C and 93·4Fe–1·6C–2·5Cu–2·5MoS_2 alloys was undertaken and the effect of process parameters on sintering behaviour was studied.[111] Components of different shapes from AISI 304 stainless steel were fabricated by Jeng *et al.*[112] using selective laser cladding technique using a Rofin Sinar RS820 1500 W CO_2 laser. The effect of process parameters on the clad height and surface quality has been studied for direct laser deposited AISI 316L stainless steel using diode laser and CW CO_2 lasers.[113,114] Dutta Majumdar *et al.*[115,116] observed that the morphology and degree of fineness of the microstructure of AISI 316L components fabricated using a 1·5 kW CW diode laser were directly related to the laser parameters. The range of grain size (maximum to minimum) was taken as a measure of homogeneity. It was found that the range of grain size and microporosity reduced to a minimum with increasing scan speed. Figure 2*a* and *b* shows the microstructures of the laser fabricated AISI 316L stainless steel with power densities of 0·031 kW mm^{-2} (Fig. 2*a*) and 0·091 kW mm^{-2} (Fig. 2*b*) and at a scan speed of 5 mm s^{-1}, with a powder feedrate of 136 mg s^{-1} (within optimum processing region).[115] It is apparent that the grain size is coarser at higher power density and the resultant morphology varied with laser parameters (mixture of cellular and dendritic at lower and predominantly coarse cellular at higher power density). The microhardness is uniform throughout the cross-section along the wall height with a marginally higher value near the substrate and a lower value at the intermediate region. The marginally higher value of microhardness near the substrate region is mainly because of the refinement of microstructure due to a high quenching rate from the underlying substrate. On the other hand, the lower level of microhardness at the intermediate region is attributed to the grain coarsening effect. Application of a higher power density decreases the average microhardness attributed to the coarsening of grains revealed from a detailed microstructural investigation.[115] Similarly, application of a higher scan speed increases the average microhardness of the fabricated layer. With increasing scan speed, due to a shorter time of interaction, a low energy is supplied during melting, resulting in the refinement of grains, and hence, an increase in the average microhardness. The effect of powder flowrate on the microhardness does not, however, show any specific trend. From the variation in microhardness with laser parameters, it may be concluded that the hardening of the formed parts is mainly because of the grain refinement and for an improved microhardness, a combination of low power and high scan speed should be chosen. Furthermore, the critical potential for pit formation E_{PP1} [550 mV(SCE)] was superior to that of the conventionally processed AISI 316L stainless steel substrate [445 mV(SCE)] also due to the pronounced uniformity and refinement of grain size.

Laser assisted rapid manufacturing of non-ferrous components

Metallic foam is increasingly becoming popular due to its unique combination of low density and high compressive strength.[121] Aluminium foam has evoked a special interest due to its light weight structure and possibilities of various structural applications in automotive, aerospace and allied industries. However, manufacturing techniques associated with aluminium foam limit the application of it in a wider scale. Kathuria[122] attempted to fabricate Al–7%Si foam using Nd:YAG laser and obtained a localised and unidirectional expansion of the foam in the direction of laser irradiation with as high as 60% porosity and closed cell structure. The synthesis involved mixing the alloy powder with a foaming agent, cold isostatic pressing of the mixture to form a foamable sandwich precursor and heating it up to its melting point by a high power laser beam. Laser irradiation is useful in inducing rapid decomposition of the foaming agent into hydrogen or some other gaseous product and allowing precipitation of intermetallic phases to strengthen the matrix phase.

Tang *et al.*[123] and Wang *et al.*[124] carried out selected laser sintering of Cu based alloy. Khaing *et al.*[125] developed a three-dimensional soft and porous structure from a mixture of Ni, bronze and Cu powders using an EOSINT M250 laser sintering machine. Kathuria[126] investigated the selected laser sintering of three-dimensional components made of Co, Ni and Cu alloys. A list of similar studies on the development of Cu and its alloys by laser assisted rapid manufacturing have been reported elsewhere.[127–129]

Ti and its alloys are widely used as aerospace components and biomedical implants. Hayano[130] and Tolochko *et al.*[131] fabricated dental root implants made of Ti by selected laser sintering. Gao *et al.*,[132] Mei *et al.*,[133] Arcella and Froes[134] and Abbott and Arcella[135] attempted the fabrication of fully dense and rapidly solidified Ti–6Al–4V components by direct laser deposition. Srivastava *et al.*[136,137] developed TiAl alloy by direct laser fabrication using gas atomised Ti48Al2Mn2Nb powder as feedstock and a VFA 600 W CO_2 laser unit. The microstructure was heterogeneous and extremely fine. A post-heat treatment was found to improve the microstructural and compositional homogeneity. Banerjee *et al.*[138–140] developed Ti–6Al–4V–TiB, Ti–25%V and Ti–TiB composites using laser engineered net shaping process. Tomochika *et al.*[141] attempted to synthesise NiTi intermetallic compound by laser assisted melting of a combination of pure Ti (300 μm diameter) and pure Ni wire (250–350 μm diameters) followed by high pressure Ar gas assisted atomisation and coating on aluminium substrate. The particle size of the atomised powder ranged from 10 to 125 μm and the thickness of the coating was 3 mm. The sintered compact of equiatomic blend of Ti and Ni was single phase NiTi martensite with martensitic M_f and austenitic finish A_f temperatures of 320 and 365 K respectively. Zhang *et al.*[142] developed TiAl intermetallics by laser engineered net shaping using a CW Nd:YAG laser. Wu and Mei[143] developed Ti and TiAl alloy parts with various geometries by direct laser fabrication. Tang *et al.*[144] prepared Ti/TiN *in situ* composites by laser induced reaction between Ti powder and N_2 gas and evaluated their mechanical properties. For complex alloy systems, the microstructure of the direct laser clad component was found to be different from the single clad layer and varied with the numbers of passes. Figure 3 shows the scanning electron micrograph of the top surface of direct laser clad Ti–8Al–1Er samples after multiclad deposition.[145] The microstructure consists of acicular α' martensite with the presence of very fine and homogeneously distributed Er rich precipitations.



3 Scanning electron micrograph of direct laser deposited Ti–8Al–1Er[145]

Karlsen *et al.*[146] developed the Co–Cr–Mo superalloy for bioimplant with improved biocompatibility. Ni and Co based superalloys were fabricated by direct laser metal deposition technique.[146–148] The tensile strength of the laser consolidated Co based superalloy was reported to be 50% higher than that of the as cast or powder fabricated components attributed to the exceptionally fine columnar dendrite microstructure produced by the rapid solidification.[147]

A new rapid prototyping method has been developed by combining selective laser sintering and gel casting technique for the fabrication of complex shaped Al_2O_3 ceramic parts.[149] The process involves aqueous gel casting to form high mechanical strength Al_2O_3 based green body by *in situ* polymerisation of Al_2O_3 slurry containing monomer and cross-linker, and sintering of the green compact by selective laser sintering machine. Larrea *et al.*[150] described the procedure for preparing large surfaces of eutectic composites of Al_2O_3–ZrO_2, with thickness up to 250 mm, using a modified laser zone melting. A ceramic precursor was prepared by compaction and sintering (at 1500°C) of powder blends with composition of 37 mol.-%ZrO_2 and 63 mol.-%Al_2O_3. The surface of a ceramic precursor is scanned with a rectangular CO_2 laser beam of $20\times0{\cdot}5$ mm size at a scan speed of 36–72 mm h^{-1}, which induces surface melting. The microstructure consisted of fine, alternating and interpenetrating Al_2O_3 and ZrO_2 single crystal lamellae.

Laser can be a useful tool for *in situ* rapid fabrication of composite components such as cutting tools, shear blades, etc. Lu *et al.*[151] fabricated TiC dispersed Cu–Ti–C and Cu–Ni–Ti–C composites by laser scanning (using a CO_2 laser) of ball milled powder mixtures of Cu–Ti–C and Cu–Ni–Ti–C. Liu and DuPont[152] developed crack free functionally graded TiC/Ti composite materials by laser engineered net shaping with compositions changing from pure Ti to ~95 vol.-%TiC. Kathuria[153] fabricated porous components of 93Cu+7Sn and 70(MoW)2NiB2:30NiMo alloy powders using Nd:YAG laser. Wang *et al.*[154] fabricated W/W_2Ni_3Si ternary *in situ* metal silicide matrix composites using direct laser deposition. Duan and Wang[155] developed rapidly solidified Cr_3Si metal silicide '*in situ*' composites by laser melting of Cr–Si–Ni alloy powders. Cheng *et al.*[156] studied the influence of process parameters on the structure and integrity of Al/SiC composite by direct laser deposition.

An intelligent laser processing based on size dependent optical absorption coefficient of metal particles has enabled fabrication of almost monodispersed metallic nanoparticles on chosen substrate.[157] Using this method, Ag clusters with mean diameters of 10 nm and size distributions of 0·13 have been prepared.[157]

Irradiation with the second harmonic generation of 1·06 μm Q switched Nd:YAG laser on powdered nonlinear crystals suspended in a photopolymeric solution enabled high resolution rapid prototyping.[158] One such structure developed by this method was a three-dimensional periodic photonic bandgap structure of aluminium oxide that consisted of layers of parallel rods forming a face centred tetragonal lattice.[159] A similar laser driven direct write deposition technique (from trimethylaminealane and oxygen precursors) was successfully utilised to fabricate a three-dimensional microstructure consisting of aluminium oxide and aluminium.[160] These laser deposited rapid prototype ceramic components are useful as micromechanical actuators such as microtweezers and micromotors.

Summary and future scope

Laser forming offers a wide range of direct or single step, contact less and novel methods of fabricating finished products/components of metallic, ceramic and polymeric origin. While bending is confined to only dimensional changes, material addition continuously, layer by layer or in sequence is apart of laser assisted direct manufacturing processes. The possibilities offered by laser forming include bending, rapid prototyping and direct manufacturing. Laser bending is attractive to both automobile and aircraft industry because of high precision and production rate achievable in this process without application of mechanical forces. Laser bending that primarily depends upon 'temperature gradient' and 'buckling' mechanisms is routinely applied in tailoring the curvature of aluminium, iron and titanium based metals and alloys. Although the process is commercially exploited in manufacturing, certain fundamental aspects need to be properly understood, particularly with regard to the precise mechanism of laser bending: Does bending necessitate partial melting at the laser irradiation zone? Why does thickness increase along the bend edge? Why does bending rate decrease with number of pass? What microstructural features (grain size, texture, etc.) accompany and affect bending? How does prior microstructural state influence the angle of bending?

Both laser assisted manufacturing and rapid prototyping are major laser forming processes that have found commercialisation in many applications. The main reasons for interest in these processes stem from the scope of direct (one step) manufacturing of round or square sections, hollow tubes and more complex geometry with the same machine and identical fixture. Solid parts are usually made in layers. Thus, the interfaces of the consecutive layers remain the weakest point. Investigations are warranted to predict the stresses generated at the edges and corners, surface roughness/contour and compositional distribution in the solid object. Development or epitaxial repair of single crystal superalloy component will be a real breakthrough. In this regard, more comprehensive treatments of heat and mass transfer in specific applications are warranted to establish the reproducibility of the laser forming processes.

Laser joining

Laser joining is a directed energy beam assisted fusion joining technique that involves the application of a high power laser beam as a source of heat to fuse and join two solids of similar or dissimilar natures. The advantages of laser joining over conventional fusion or arc welding processes include high welding speed, narrow heat affected zone (HAZ), low distortion, ease of automation, ability for single pass joining of thick sections and better design flexibility with controlled bead size.[2] The special interest in laser joining lies in its capability of fusion joining of dissimilar materials with different section thicknesses, compositions and/or physical/chemical properties for industry scale applications particularly in automotive and aerospace industry.[161–168] Laser joining encompasses welding, brazing, soldering and microwelding. Joining of materials requires a laser source capable of delivering high power density with precision, reliability, flexibility and overall economy. Hence, pulsed or CW Nd:YAG or CO_2 laser (very seldom ruby laser) and recently the diode lasers are the commonly used lasers for joining. The main process variables in laser assisted joining are laser power, beam diameter, beam configuration, travel speed of the workpiece, substrate condition (roughness and temperature), filler type/feedrate, alloy composition and thermophysical properties (melting temperature, specific heat, density and thermal conductivity) of the workpiece. Table 2 presents a ready reference for laser assisted joining of iron based and

Table 2 Summary of selected recent studies on laser joining of materials

a. Laser assisted welding of ferrous metals and alloys

Scope/material	Objectives/outcome	Reference	Year
AISI 304 stainless steel	Fatigue behaviour of laser welded specimens in air, hydrogen and Ar using a CW CO_2 laser	Tsay *et al.*[228]	2003
Austenitic and duplex stainless steel	Study of microstructures of the welded zone using Nd:YAG laser	McPherson *et al.*[221]	2003
Duplex stainless steel 22Cr–5Ni–3Mo (UNS S32205)	Optimum conditions to minimise the δ-ferrite content in weld zone formed by CO_2 laser	Capello *et al.*[229]	2003
AISI 304 stainless steel	A finite element based heat transfer model to predict the weld bead shape and thermal distortion using Nd:YAG laser	Chang and Na[230]	2002
Ni+Au–Ni plated steel	Effect of Au/Ni plating on strength of welded steel sheets using pulsed Nd:YAG laser	Biro *et al.*[172]	2001
Steel	Effect of temperature dependence of surface tension on welding of steel using CW laser	Fuhrich *et al.*[173]	2001
AISI 304 stainless steel	Extremely narrow welding seam with fully austenitic microstructure in laser welded AISI 304 stainless steel using photolytic iodine laser	Farid and Molian[174]	2000
Galvanised steel	Relation between defects in welding with weld speed, average power density and pulse duration	Tzeng[231]	2000
High carbon steel	A crack free, stronger and wider weld pool by CO_2 laser welding	Ng and Watson[175]	1999
Plain C and AISI 304 stainless steel	Welding by feeding of metallic wire	Sun *et al.*[224]	1998
Low carbon galvanised steel	Effect of shielding gas on weld penetration and formability	Chung *et al.*[218]	1999
Mild steel	Strength of welded joint made by laser spot welding and resistance welding	Yang and Lee[216]	1999
AISI 304 stainless steel and free cutting steel	Laser welding for fabrication of hydraulic valves	Li and Fontana[232]	1998
AISI 304 stainless steel	Combination of pulsed beam with high peak power and modulated CW beam to obtain deep penetration	Narikiyo *et al.*[233]	1998
High strength low alloy steel	Fatigue behaviour and cracks in laser welded steel	Onoro and Ranninger[234]	1997
15–5PH (0·04C–14·8Cr–4·77Ni–3·29Cu–0·73Mn–0·53Si)	Non-equilibrium structure and phase analysis	Kusuda *et al.*[235]	1997
Austenitic stainless steel	Influence of high speed and low power on welding by CW CO_2 laser	El-Batahgy[176]	1997
Austenitic stainless steel	Relation between weld zone size and shape with applied power and scan speed	El-Batahgy[222]	1997
Stainless steel and titanium	Plasma plume temperature and equilibrium	Szymanski *et al.*[177]	1997
Stainless less, Kovar, gold	Influence of power density on defect/hole and centreline cracks	Cheng *et al.*[178]	1996
Stainless steel	Effect of gravity on Cr distribution	Wang and Tandon[179]	1995
Tritiated AISI 316L stainless steel	Segregation of tritium in the weld and HAZ	Roustila *et al.*[236]	1994
Electrogalvanised steel sheet	Formation of Zn and Fe rich intermetallics in the weld zone	Akhter *et al.*[237]	1990

b. Laser assisted welding of non-ferrous metals and alloys

Laser welding of Al/Al alloys

Material	Objectives/outcome	Reference	Year
5182 and 5752 Al alloys	A model to predict fusion zone geometry for keyhole mode of LBW with experimental validation using a CW Nd:YAG laser	Zhao and DebRoy[243]	2003
AA 5083-O and A356-T6 aluminium alloys	Effect of single beam and dual beam welding on the microhardness and composition	Haboudou *et al.*[247]	2003
Al	Effect of application of external electrical current on laser welding using a TLF3000 CO_2 laser and fluid flow	Xiao *et al.*[187]	2001
C95800 Ni–Al bronze	Effect of welding speed on weld quality and microstructure during diode pumped Nd:YAG laser assisted welding	Leong *et al.*[167]	2000
6061-T6 Al–Mg–Si alloy	A kinetic analysis for dissolution of Mg_2Si during laser welding using a 2·5 kW CW CO_2 laser	Hirose *et al.*[168]	1999
AA 1100 Al alloy	Conditions for pore free, conduction pool geometry welding during pulsed Nd:YAG laser	Weckman *et al.*[169]	1997
SiC dispersed Al–Cu–Mg based metal matrix composite	Carbide formation kinetics with respect to laser intensity and pulse duration during welding using a pulsed Nd:YAG laser	Yue *et al.*[249]	1997
8090 Al–Li alloy	Studies on fusion zone dimension, solute evaporation, hardness, porosity and tensile properties of the weld zone welded using 3 kW CW CO_2 laser	Lee *et al.*[170]	1996
Al–Fe–V–Si alloy	Formation of Si/Fe/V rich Al_4Fe type precipitates and Si/Fe rich phase on boundaries of CW CO_2 laser welded alloy	Whitaker and McCartney[171]	1995

Table 2 Continued

Material	Objectives/outcome	Reference	Year
Laser welding of Ti based materials			
Ti–24Al–17Nb	Studies on weld microstructure of CO_2 laser welded sample	Wu *et al.*[253]	2002
TiNi shape memory alloys	Influence of laser parameters on M_s temperature and ductility, amount of β_2 phase and strength related to s hape memory effect	Hsu *et al.*[180]	2001
Ti–6Al–4V alloy	Influence of power density on keyhole formation and melt depth during pulsed Nd:YAG laser welding	Perret *et al.*[181]	2001
Ti	Comparison of electron beam, laser and TIG welded Ti	Qi *et al.*[251]	2000
Ti–6Al–4V and Ti–6Al–2Sn–4Zr–2Mo	Butt joint (using filler wire) with a CW CO_2 laser and autogenous welding by pulsed Nd:YAG laser	Li *et al.*[255]	1997
Ti–6Al–4V	Crack initiating mechanism due to laser welding using a CW CO_2 laser	Tsay and Tsay[182]	1997
TiAl	Joining by self-propagating high temperature synthesis reaction using a 2·5 kW CW CO_2 laser	Uenishi and Kobayashi[254]	1996
SiC–fibre reinforced Ti alloy	Influence of scarf angle on butt and scarf joints using a CW CO_2 laser	Hirose *et al.*[183]	1995

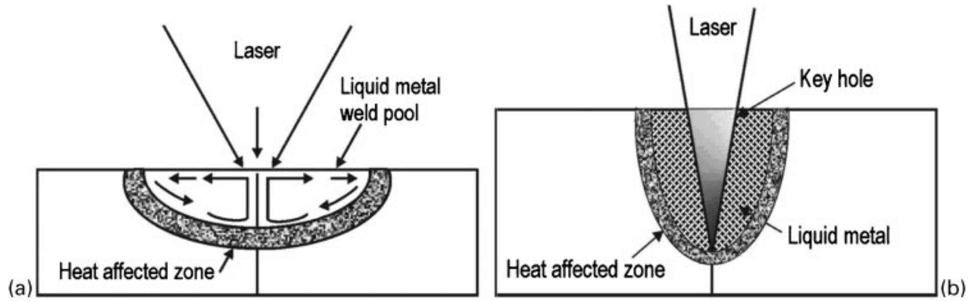
The welding efficiency can be expressed in terms of the power (or energy) transfer coefficient η, where η is the ratio between the laser power absorbed by the workpiece and the incident laser power. η is usually very small but can approach unity once a keyhole has been established. The melting efficiency or melting ratio ε is given by

$$\varepsilon = \frac{vdW\Delta H_m}{P} \tag{1}$$

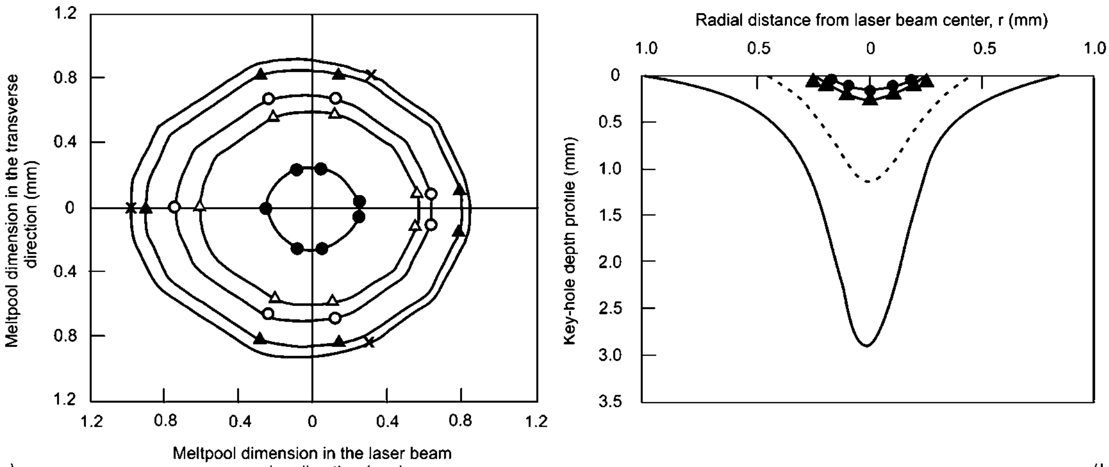
where ε is the rate of melting, P is the incident laser power, v is the welding speed, d is the sheet thickness, W is the beam width and ΔH_m is the enthalpy or heat content of the metal at the melting temperature. The maximum values of ε are 0·48 and 0·37 for penetration welds and conduction welds respectively.[166]

It is apparent that ε never approaches unity even when $\eta \cong 1$. Both η and ε can be enhanced under the keyhole welding condition if the absorption coefficient can be increased. This can be accomplished by application of absorbent coating, using smaller wavelength laser light, surface roughening or texturing, preheating, tailoring of temporal irradiation profile and/or oxidation/nitriding. Among various process parameters, the quality and properties of laser weld depend on laser pulse time and power density, laser spot diameter/penetration, traverse speed of the laser beam across the substrate surface, shielding gas, melt area, melting ratio and material properties such as absorptivity, specific heat, density, etc. For details about the effect of process parameters on laser welding, the reader may refer to Duley.[166]

The combination of laser beam with metal inert gas or tungsten inert gas (TIG) arc (so called hybrid technique) seems promising from the viewpoint of bead formation except that occurrence of large blowholes and voids can pose serious problems. Recently, Kern *et al.*[186] have detected an intrinsic current in the melt pool driven by thermoelectric potential. It is shown that the application of external magnetic field in laser welding of steel can modify the shape of seam cross-section, level of porosity and bead geometry at higher speed of welding. Xiao *et al.*[187] modified the magnetofluiddynamic approach by applying an electrical current through an external power source through a 2·4 mm diameter tungsten electrode during laser welding using a TLF5000 CO_2 laser and a TLC 100 five axis laser processing machine. Bead on plate welds were made on 5 mm thick plates of Al. Helium with a flowrate of 25 L min^{-1} was supplied by a side pipe with an inner diameter of 6 mm to suppress formation of laser induced plasma and protect the weld pool from the atmosphere. The electrical current flowing in the weld pool induced an azimuthal magnetic field that was proportional to the current density resulting in electromagnetic forces proportional to the square of current density. Because of the divergence of the current lines from the electrode tip to the workpiece, the current density distribution in the weld pool turned non-uniform such that the current density close to the electrode tip and keyhole front was larger than that at the bottom and rear of the melt pool. Accordingly, the molten metal in front of the keyhole was accelerated towards the



4 Schematic of *a* conduction melt pool (semicircular) and *b* deep penetration (keyhole) fusion zone and weld[166]



(a) (b)

5 *a* melt pool size and shape at substrate surface under quasi-steady state condition and *b* depth of keyholes formed during laser welding for incident laser power P=1850 W, laser beam scanning speed v=2·12 cm s^{-1} and absorptivity A=8% (Ref. 197)

bottom that produced additional heat to the region. The weld depth, therefore, increased, while the width decreased. It was concluded that an external current can significantly influence the fluid flow of the weld pool and shape of the seam cross-section in laser welding of aluminium, to result in improved flexibility.

Hirsch *et al.*[188] developed a finite element model of heat and mass transfer to predict the weld pool geometry and weld depth during CO_2 laser welding of ceramics and compared the predicted result with the experimentally observed values. The predicted weld pool shape agreed well with the experimentally observed one. Several investigations address similar issues of modelling the heat/mass transfer and fluid flow during laser welding to predict thermal history, shape and size of the weld pool.[187–195]

During deep penetration laser welding, the plasma created over the keyhole absorbs beam energy and reduces power efficiency. Thermal movement of laser induced plasma including the principle and feasibility of controlling the plasma by electric and magnetic fields was theoretically and experimentally analysed by Peng *et al.*[196] An experimental procedure involving raising the nozzle during laser welding was used to evaluate the effect of increasing the power efficiency by driving away the charged particles. The power efficiency increased with increasing magnetic field intensity. The power efficiency appears to reach the highest value at an optimal electric and magnetic field intensity.

Kar and Mazumder[197] developed a model to describe the laser keyhole welding to determine the free surface velocity and temperature distributions as well as the shape of the keyhole. The velocity and temperature fields in the molten material can be obtained by considering the following momentum, energy and mass conservation equations

$$\rho l\left[\frac{\delta u_l}{\delta t}+(u_l\nabla)u_l\right]=\rho_l g-\nabla p_l+\mu_l\nabla^2 u_l \qquad (2)$$

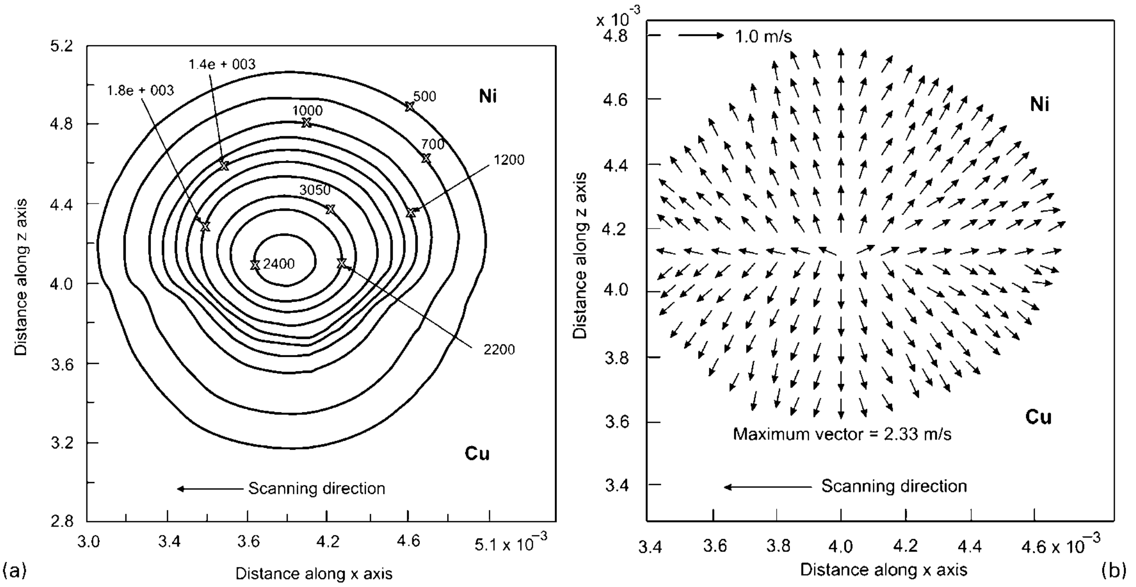
$$\rho_l c_{pl}\left[\frac{\delta T_l}{\delta t}+(u_l\nabla)T_l\right]=k_l\nabla^2 T_l \qquad (3)$$

$$\nabla u_l=0 \qquad (4)$$

where ρ_l, μ_l, c_{pl}, k_l, u_l, g, p_l and T_l are the density, viscosity, specific heat at constant pressure, thermal conductivity, velocity vector, acceleration due to gravity, pressure and temperature of the molten material respectively. t and ∇ are the time variable and differential operator respectively.

Figure 5*a* shows the shape of the weld pool at the top surface of the workpiece. The intersection of the vertical and horizontal lines in the middle of this figure indicates the centre of the laser beam which moves to the right along the horizontal line. The innermost curve in this figure is the liquid/vapour interface which represents the size of the mouth of the keyhole at the substrate surface. The mouth of the keyhole is found to be almost similar for the welding parameters used for deriving the weld pool geometry (Fig. 5*a*). However, the outer four curves are the solid/liquid interfaces representing the sizes of the melt pool at the substrate surface, indicating that weld pools of different sizes can be obtained for different welding conditions. These four curves also show that the leading solid/liquid interface is closer to the centre of the laser beam than the trailing solid/liquid interface, which is behind the laser beam opposite to the scanning direction. This effect arises because continuous heating occurs ahead of the laser beam along the direction of scanning while cooling takes place behind the beam in the opposite direction. Figure 5*b* has been obtained by plotting the solid/liquid and liquid/vapour interfaces on the vertical plane that coincides with the laser beam scanning direction and indicates that the depths increase with increasing welding time until the quasi-steady state is reached. Furthermore, the depth of the melt pool increases more rapidly than the depth of the keyhole as the welding time increases.

Rai *et al.*[198] presented a convective heat transfer model to calculate temperature and fluid flow fields during keyhole mode laser welding for different mechanisms of heat transfer in the weld pool. A vorticity based turbulence model has been used to estimate the values of effective viscosity and thermal conductivity in the melt. The model is used to calculate the temperature and velocity fields, weld pool geometry, cooling rate and solidification parameters for welding of AISI 304 stainless steel and Al 5754 aluminium alloy.



6 ***a*** **isotherms along the top surface of laser welded Cu with Ni after 25 ms from turbulent simulation (temperatures are shown in °C). In the contour labels, 1×10^3 indicates 1000 units.** ***b*** **top surface velocity vector in the molten pool after 25 ms in turbulent simulation. In moving coordinate system laser torch located at *x*=4 mm and *z*=3·94 mm. At time *t*=0 s, *z*=3·94 mm indicates the separation line between Cu and Ni workpieces[200]**

The morphology of the solidification front during welding may be predicted from the ratio of thermal gradient G to solidification rate R_s. The criterion for plane front instability based on constitutional supercooling is given by the following relation[199]

$$G/R_S < \Delta T_E / D_L \qquad (5)$$

where ΔT_E represents the temperature difference between the solidus and liquidus temperatures of the alloy and D_L is the diffusivity of a solute in the liquid weld metal. $\Delta T_E/D_L$ for aluminium alloy is found to be of the order of 10^3. Figure 5*b* shows the variation in G/R_s with net energy input for aluminium alloy. It is observed that G/R_s increases with increasing net energy input, which indicates that the solidification microstructure will become less dendritic with decreasing welding velocity.

Computational modelling of welding of dissimilar metals has been extensively studied by several investigators.[200,201] The effect of turbulence on momentum, heat and mass transfer during laser welding of a copper–nickel dissimilar couple has been studied by carrying out three-dimensional unsteady Reynolds averaged Navier–Stokes simulations.[200] Figure 6*a* shows the isotherms on the top surface, as obtained from turbulent simulations showing the shifting of the maximum temperature location towards the nickel side. A combined consequence of the differences in thermal diffusivities of Cu and Ni and their melting temperature causes the convective transport of molten pool in a complex manner. The velocity vector plots for turbulent simulation of the molten metals are depicted in Fig. 6*b*, where it is evident that in the case of turbulent transport, the mean thermal and concentration gradients decrease due to an increased diffusion strength.

This decreases the strength of Marangoni convection in turbulent flow eventually. Furthermore, a greater degree of momentum diffusion in the turbulent pool results in a decrease in the maximum velocity magnitude, as is evident from Fig. 6*b*. As a result, the mean advection of heat from the centre of the pool towards its peripheries is weaker than that in the case of laminar transport, which tends to decrease the pool width. The effects of turbulence are found to be most prominent on the species transport in the molten pool. The composition distribution in turbulent simulation is found to be more uniform than that obtained in the simulation without turbulent transport. The nickel composition distribution predictions, as obtained from the present turbulence model based simulations, are also found to be in good agreement with the corresponding experimental results.

A large variety of metals and alloys have been welded with pulsed or CW lasers. The earliest study on deep penetration laser beam welding (LBW) was earlier outlined by Brown and Banas.[202] The potential scope of the application of pulsed laser welding was earlier outlined by Duley,[203] Ready,[204] Berretta *et al.*[205] and Schwartz.[206] Li and Gobbi[207] and Schubert *et al.*[208] gave a brief overview on the potential applications of high power CO_2 and Nd:YAG lasers to create lightweight structures in the automotive, aerospace and aeronautic industries. In this context, the welding behaviours of Ti–6Al–4V (for aeronautic application), Inconel 718 (exhaust frame) and Hastelloy X (combustion liner) have been discussed.

Owing to a large thermal gradient associated with laser assisted welding, the generation of residual stress leads to the formation of distortions and defects in the welded joint. Postacioglu *et al.*[209,210] predicted the distortion generated in a weld pool using linear thermoelastic theory. Niordson[211] analysed the fracture toughness of steel joined by a laser welding in terms of steady state crack growth along the weld. Laser welding gives a high mismatch of yield strength between the weld zone and base material, due to the fast thermal cycle that the material undergoes in welding. Fracture is modelled by using a cohesive zone criterion in front of the crack tip along the fracture zone. It is found that a thinner laser weld gives higher interface toughness. Furthermore, it is shown that the preferred path of the crack is in the base material slightly outside the weld, a phenomenon also observed in experiments.

In the following subsections, a brief overview of the studies on laser assisted welding of specific ferrous and non-ferrous alloys has been discussed.

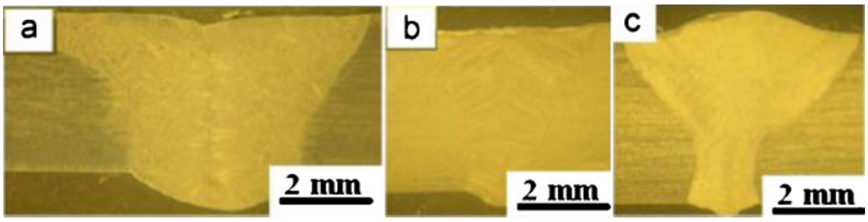
Laser assisted welding of Fe based materials

Laser assisted welding has been successfully implemented in a large number of ferrous alloys.[212–237] Baarden *et al.*[212] carried out laser welding of rimmed steel for automotive application. Laser assisted welding of several grades of stainless steel was attempted for applications in power plant and chemical industries.[213,214] High speed laser welding of tin plate and tin free steel for container industries was reported by Mazumder and Steen.[215] Until recently, automobile manufacturing extensively used resistance spot welding. However, resistance spot welding requires many welding machines for parts with geometrical and structural complexity and hence, is now substituted with laser spot welding, which seems advantageous over the former technique both in terms of low cycle fatigue strength and residual stress.[216] Yang and Lee[216] measured the low cycle fatigue strength and residual stress of laser spot welded mild steel (using a CW CO_2 laser) and compared the weld strength with that welded by resistance spot welding. Finite element modelling revealed that the residual tensile stress was distributed over the inner and outer rims of the fusion zone. Banas[217] conducted high power laser welding experiments with austenitic stainless steel and obtained a dense and defect free weld zone, as revealed by radiographic inspection. The welded joint strength was equivalent to that of the base metal. Chung *et al.*[218] evaluated the influence of shielding gas (helium, carbon dioxide, argon, nitrogen and 50% argon+50% nitrogen) on CO_2 laser weldability of low carbon 1·5 mm thick automotive galvanised steel. Helium and 50% argon+50% nitrogen were the best shielding gases while pure argon as shielding gas was most effective in terms of penetration and formability for welding with high power CO_2 laser welding. The shielding gas determines the period, size and shape of the plasma created by laser irradiation depending on the ionisation potential, dissociation potential, thermal conductivity and atomic weight of shielding gas. Malek Ghaini *et al.*[219] studied the influence of laser pulse energy, duration and travel speed on weld dimension, microstructure and hardness during overlapping laser assisted bead on plate spot welding of low carbon steel using a pulsed Nd:YAG laser. It was found that significant variations in the microstructure and hardness of weld within a weld spot could arise in full penetration welding due to the rapid solidification process associated with the technique.

Kwok *et al.*[220] fabricated austenitic (S30400 and S31603), duplex (S31803) and super duplex (S32760) stainless steels by laser penetration welding with a CW Nd:YAG laser in an argon atmosphere. While the laser welded S31603 retained the original austenitic structure, the laser welding of S30400 contained austenite as the major phase and δ-ferrite as the minor phase. On the other hand, a slight change of δ-ferrite to austenite ratio was found in both the laser welds of S31803 and S32760, with austenite present at the δ-ferrite grain boundaries. The welds exhibited passivity but their pitting corrosion resistance deteriorated as compared with that of the unwelded specimens. The decrease in pitting corrosion resistance of the welds was attributed to the microsegregation in the weld zone of S31603, and to the presence of δ-ferrite in S30400. For the duplex grades S31803 and S32760, the change of the ferrite/austenite phase balance in the weld zone could be the cause of the decrease in corrosion resistance.

McPherson *et al.*[221] achieved autogenous welding of austenitic stainless steel and duplex steel using an Nd:YAG laser and compared the results with that obtained by submerged arc welding. Laser welding produced higher hardness in the weld zone and prevented distortion. El-Batahgy[222] studied the welding behaviour of AISI 304L, 316L and 347 stainless steels with different sheet thicknesses. The fusion zone shape and final solidification microstructure were evaluated as a function of laser parameters. Both bead on plate and autogenous butt weld joints were made using a carbon dioxide laser with a maximum output power of 5 kW in CW mode. Depending on the base metal thickness, parameters such as laser power, welding speed, defocusing distance and shielding gas should be carefully selected so that weld joints having complete penetration, minimum fusion zone size and acceptable weld profile are produced. Heat input as a function of both laser power and welding speed has almost no effect on both the type of microstructure and mechanical properties of welds. The microstructure of all laser welds was always austenitic with about 2–3% δ-ferrite. However, the lower the laser power and/or the higher the welding speed, the finer the solidification structure. Primary ferrite or mixed mode solidification resulted in crack free welds. Chan *et al.*[223] carried out welding of 0·5, 0·6, 0·8 and 1·0 mm thick steel sheets to form tailored weld blanks with different thickness ratios of 2 (0·5/1·0 mm), 1·67 (0·6/1·0 mm) and 1·25 (0·8/1·0 mm) using a Nd:YAG laser. Subsequent formability and uniaxial tensile tests showed that tailored weld blanks of different dimensions and cutoff radii yielded different major and minor strain values. Furthermore, the forming limit diagrams of the tailor welded blanks indicated that both forming limit and minimum major strain value decreased as the thickness ratio would increase. This implied that the higher the thickness ratio, the lower the formability of the tailored weld blanks.[223]

The welding of low alloy 13CrMo44 steel to AISI 347 stainless steel was carried out using 1·2 mm diameter ENiCrMo-3 nickel based wire as filler.[224] The joints were in the tube form with an outer diameter of 43·5 mm and a wall thickness of 4·5 mm. Mixing behaviour of the filler with the base metal was evaluated by measuring Cr and Ni contents in the weld metal. Composition along the weld zone was uniform producing uniform hardness throughout. Costa *et al.*[225] studied the weldability of hard metal (K40, 88%WC–12%Co and K10, 94·4%WC–5·6%Co) to steel with a high power CW CO_2 laser or Nd:YAG laser and pulsed Nd:YAG laser. Weld bead size, microstructure, bending strength and hardness were evaluated to correlate the mechanical properties with the concerned welding parameters to obtain full penetration and high strength joints. K40 (88%WC–12%Co) quality tips have higher resistance after welding, compared with that in K10 samples, that contained cracks after welding. It appears that diffusion of tungsten to the bead during welding leads to the formation of several carbides in the martensitic matrix and induces higher bead hardness. In spite of having eutectic structure with multiple carbides, the weld bead did not develop cracks. It was concluded



7 Macrostructure of weld zone formed by *a* TIG welding, *b* laser welding and *c* LATIG hybrid welding techniques[240]

that laser welding particularly with CW Nd:YAG laser was an effective joining technique for hard metals, *vis-à-vis* brazing or mechanical clamping, for manufacturing cutting tools with small weld beads and HAZ and a minimum residual stress.

Electrogalvanised steel sheet with zinc coating (thickness of 7·5 μm) was laser welded using a CW CO_2 laser with 23 mm beam diameter having near Gaussian beam configuration. Subsequent electrochemical tests in 1 N NaCl solution revealed the presence of Zn–Fe intermetallic phases in the HAZ that accounted for the improved adhesion and better sacrificial protection against corrosion.[226] Iqbal *et al.*[227] developed a new method of using two tandem laser beams for lap welding of galvanised steel sheets. This involves a precursor beam and a higher power actual beam, generated independently or otherwise split from the same source. The first beam cuts a slot, thus making an exit path for the zinc vapours, while the second beam performs the desired welding. The process is easier as it does not need any preprocessing, prearranging with additional components or contour limitations and the welders may proceed with the jobs as per non-coated steel and applying the technique, the weld zone can get rid of all or a part of the zinc vapour porosity at lower or higher speed respectively. The experimental results seem promising showing total absence of zinc in the welds. However, the selection of proper source, beam alignment (between weld and cut) and proper sheet clamping are important.

Hybrid laser welding is a technique which combines both the laser welding and an arc welding process, to weld materials. By application of the technique, the beneficial effects of both the techniques are achieved. Based on the combinations of specific laser source and other welding sources, there are a number of hybrid laser welding. The advantages of laser hybrid welding over the individual welding technique include flexibility in welding of a thicker section, a high welding speed, less distortion, less weld defects and flexibility of bridging of a larger gap.[238,239] A detailed study of the microstructures of welded zones developed by TIG welding, laser welding and laser–TIG (LATIG) hybrid welding was undertaken by Yan *et al.*[240] It was observed that all the welding techniques were capable of producing full penetration joints. Figure 7 shows the macrostructures of the weld zone. From Fig. 7, it is evident that the zone is free from cracks and porosities. The shape and size of the weld zone were, however, different in three cases. Different widths and fusion areas of different weld joints are attributed to the different welding speeds and heat inputs used in the different welding methods. Figure 8 shows the microstructures of the fusion zones of the three joints at high magnification. All joints are composed of dark δ-Fe dendritic structure in austenite matrix. However, different dendrite sizes can be observed in the three joints. The δ-Fe dendrite size in TIG joint is obviously larger than that in laser and hybrid joints. The primary and secondary dendrite arm spacings of TIG joint are about 10–14 and 3–7 mm respectively. The laser joint shows the smallest dendrite size in all joints. The primary dendrite arm spacings of laser and hybrid joints are about 2–5 and 4–8 mm respectively. Furthermore, the δ-Fe in laser hybrid joint exists as network morphology in fusion zone, while exists as columnar structure in laser joint.

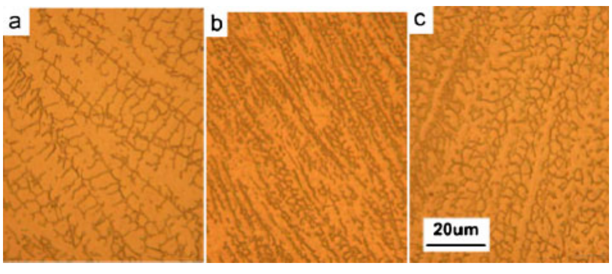
The tensile strengths of TIG, laser and hybrid joints are 560, 733 and 683 MPa respectively. The TIG joint exists as cup–cone shaped fracture, while the laser and hybrid joints exist as pure shear fracture. The laser welding and hybrid welding are suitable for welding 304 stainless steel in industry application owing to the high welding speed and excellent mechanical properties.

Gao *et al.*[241] studied the microstructural characteristics of mild steel weld joints developed by CO_2 laser–metal inert gas hybrid welding.

Laser assisted welding of non-ferrous metals

Compared with the number of studies on welding of titanium, plain carbon and alloy steels using high power lasers, the welding of aluminium did not attract the same level of research attention mainly because aluminium alloys pose several difficulties in welding due to the high surface reflectivity of laser and affinity for oxygen. Nevertheless, the scope and relevant issues of laser assisted welding of lightweight metals such as Al, Mg, Ti and their alloys for aerospace applications have been reviewed by Schubart *et al.*[208]

Both conduction mode welding and keyhole mode welding are possible in aluminium.[242] This porosity problem can be reasonably eliminated by surface milling and vacuum annealing.[242] Zhao and DebRoy[243] have developed a model to predict the fusion zone geometry for keyhole mode LBW of 5182 and 5752 Al alloys using a CW Nd:YAG laser and compared the results with



8 Microstructure (at high magnification) of weld zone formed by *a* TIG welding, *b* laser welding and *c* LATIG hybrid welding techniques[240]

experimental results. Dousinger *et al.*[244] described the factors relevant to deep penetration welding, process efficiency and process stability for welding of aluminium alloys.

Autogenous 'bead on plate' laser beam welding of Al alloys by a 3 kW CO_2 laser under Ar or N_2 atmosphere is possible in the 700–1300 W power range and 1500–9000 mm min^{-1} scan speed with the focus located at 1–3 mm below the surface.[245] Though bead on plate experiments do not represent a true joining exercise, these studies are helpful to correlate the dimension and geometry of melt pool and HAZ with laser and material parameters. The effects of different gases were evaluated in terms of weld line appearance, fusion zone dimension, solute evaporation, microhardness, post-weld tensile properties and porosity distribution. In comparison with the electron beam welding, laser welding yielded a higher fusion zone depth to width ratio, cooling rate and porosity, and a lower solute loss and post-weld tensile strain.[245] A similar investigation on the microstructural evaluation following autogenous bead on plate CO_2 laser welding of an Al–8·5Fe–1·2V–1·7Si alloy (wt-%) on 2 mm thick sheet showed that the fusion zone microstructure consisted of faceted precipitates of $\sim$10 μm embedded in cellular–dendritic α-Al matrix with submicrometre intercellular phases.[246]

Continuous wave Nd:YAG laser welding of AA 5083-O wrought aluminium alloy with a high Mg content (4·5%) and A356 cast aluminium alloy with 7%Si and a cast oxide layer was carried out by Haboudou *et al.*[247] It was concluded that surface preparation as well as dual beam welding are adequate for reducing porosity in laser assemblies. Braun[248] did a detailed study on butt welding of a 6013 aluminium alloy sheet using a 3 kW Nd:YAG laser with two kinds of filler metals: gas atomised powders of aluminium alloys AlMg5, AlSi12, AlSi12Mg5 and AlSi10Mg, as well as powder mixtures of Al and Si and the binary alloys Al–5Mg, Al–5Zr, Al–5Cr and Al–10Mn. The weld region exhibited a dendritic/cellular structure in the fusion zone and a partially melted zone beside the fusion boundaries. The effect of filler wire composition and post-weld treatment on the microstructure, hardness and corrosion behaviour was established to optimise the process parameters. Yue *et al.*[249] studied the welding behaviour of a SiC particulate reinforced 2124 Al alloy composite using a pulsed Nd:YAG laser. The influences of laser parameters, i.e. laser intensity, pulse duration and focus position on the depth and size of fusion zone were investigated. Tendency to crack formation could be minimised at an applied power density of $<4\times10^9$ W m^{-2} and a pulse duration of 8–10 ms. Wang *et al.*[250] have demonstrated the versatility of laser welding by carrying out *in situ* weld alloying and LBW to join SiC reinforced 6061 Al metal matrix composite using titanium as filler alloying element. Microstructural studies show that the detrimental needle-like aluminium carbides are completely eliminated and the central fusion joint consists of TiC, Ti_5Si_3 and Al_3Ti along with some large pores.

A comparative study of the microstructures and properties of 0·5 mm thick welded sheets of commercial purity titanium by high vacuum electron beam welding, CO_2 LBW and TIG welding processes showed that the electron beam welding is more suitable for defect free welding of commercial purity titanium.[251] The mechanism of keyhole formation during pulsed Nd:YAG laser interaction with Ti–6Al–4V alloy was established by Perret *et al.*[252]

Wu *et al.*[253] welded a Ti–24Al–17Nb (at-%) alloy with a fast axial flow type CO_2 laser of 3·0 kW rated power. The weld microstructure consisted primarily of ordered β phase (namely β2 phase) and was independent of the laser parameters. The tensile strength and ductility of the joints were not affected. Uenishi and Kobayashi[254] attempted fusion welding of Ti–46Al–2Mo using a 2·5 kW CW CO_2 laser and established the parameters for crack free welding. A detailed study of laser welding of Ti–6Al–4V and Ti–6Al–2Sn–4Zr–2Mo sheets using CO_2 and Nd:YAG lasers showed that the porosity level could be minimised by CO_2 welding with filler wire.[255] In this regard, it is relevant to note that marginal deteriorations in tensile strength and fracture toughness were observed in laser welded of Ni–51 at-%Ti alloys using Nd:YAG laser.[256]

Spot welding of Ta plates was performed with a pulsed Nd:YAG laser (50 W) coupled with a step index fibre. A series of spot welds were achieved with a peak power of 3 kW (i.e. a power density of 42×10^9 W m^{-2}) and pulse duration between 1 and 16 ms.[257] The presence of small and large keyholes suggested that the keyhole geometry and solidification time primarily determined whether volume defects (porosity) would form and optimisation of process parameters could minimise the tendency of defect formation.

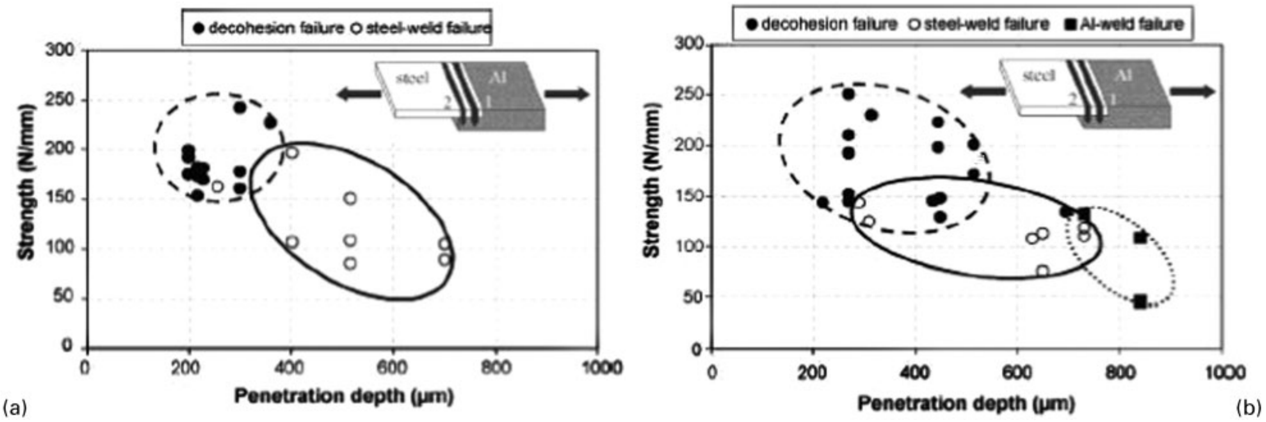
The optimum process parameters during pulsed laser welding (using 1·6 kW pulsed YAG laser) of V–4Cr–4Ti alloy was determined.[258] Laser beam welding of Hastelloy X was performed using a 1·2 and 6 kW CO_2 laser with lap and butt joints.[259] Cracks in the fusion zone were observed which could be eliminated by changing the beam diameter and alignment.

Laser welding process was introduced into dentistry by the end of 1980s causing a great impact on development of cheaper and smaller equipment using simpler technique. Ag–Pd–Au–Cu alloy used for dental implants was welded using a Dentaurum DL 20002S Nd:YAG laser at a beam power of 6·08 kW and an interaction time of 14 ms, producing a welding energy of $\sim$85·12 J, leading to the formation of refined microstructure with a high corrosion rate.[260]

Song *et al.*[261] studied the weldability of AZ31B magnesium alloy sheets by laser–arc hybrid welding process. A sound overlap welded joint of magnesium alloy AZ31B sheets without defects was obtained. Liu *et al.*[262] compared the weldability of AZ31B magnesium alloy using hybrid LATIG welding, LBW and TIG welding and found that the welding speed of LATIG was higher than that of TIG but comparable with that of LBW, and the penetration of LATIG was twice that of TIG and four times that of LBW. In addition, arc stability was better in hybrid LATIG welding than TIG welding, especially at high welding speed and low TIG current. It was found that the HAZ of joint was only observed in TIG welding, and the size of the grains was evidently coarse.

Kim *et al.*[263] studied the welding of $Cu_{54}Ni_6Zr_{22}Ti_{18}$ metallic glass using pulsed Nd:YAG laser. It was observed that crystallisation could be suppressed both in weldment and HAZ at a welding speed of 60 mm min^{-1}.

Laser microwelding of $AlMg_3$ with a thickness of 0·2 mm and AlMg4·5 Mg foils of 1 mm in thickness was



9 Linear transverse tensile strength versus penetration depth for joints welded with *a* 150 mm length focal lens and *b* 200 mm focal lens. The presence of three failure mode may be noted: decohesion mode with failure on the weld/aluminium interface of the two joints, a weld and steel/weld interface failure mode in joint 2 and an aluminium/weld interface failure in the lower part of the joint 1 (Ref. 267)

conducted using a pulsed Nd:YAG laser and the effects of pulse shaping on the welding shape and microstructures were investigated experimentally with numerical studies.[264]

Laser assisted welding of dissimilar materials

Joints between dissimilar metals are commonly encountered in the power generation, chemical/petrochemical plants, nuclear and electronics industries for the purpose of tailoring component properties or weight reduction. More recently, laser welding technologies have been successfully used to manufacture hybrid microsystems consisting of different materials.[265] Sun and Ion[266] presented a brief overview of laser welding of dissimilar metal combinations, the theory behind the weldability materials combination and highlighted the above benefits in a number of practical applications.

Sierra *et al.*[267] studied the weldability of low carbon steel to 6000 series aluminium alloy in keyhole welding mode using a CW Nd:YAG laser. Defect free welding can be obtained on overlapping aluminium and steel joints with steel on aluminium with the weld zone composed of solid solution of aluminium in iron and FeAl phases when the penetration of steel in aluminium was below 500 μm. Embrittlement at the joining zone was observed, mainly located on the weld/aluminium interfaces comprising Fe_2Al_5 and/or $FeAl_3$ phases with thicknesses between 5 and 20 μm. Figure 9*a* and *b* shows the variation in linear failure strength (N mm^{-1}) which corresponds to the strength divided by specimen width as a function of penetration depth for 150 and 200 mm focal lengths. It is apparent that the general strength decreases with increasing penetration with a maximum strength close to 250 N mm^{-1} obtained for 200–400 μm penetration depths. Increasing penetration depth leads to a change in the assembly and creation of weak points (in the weld and on the steel/weld interfaces) and induces a severe decrease in strength. Welds obtained in an aluminium on steel configuration always exhibited cracks at the interfaces.

Majumdar *et al.*[268] developed a crack free weld between a commercially available Ti alloy (Ti–6 wt-%Al–4 wt-%V) and a wrought Al alloy (Al–1 wt-%Mg–0·9 wt-%Si). Intermetallic compounds (mainly TiAl and Ti_3Al) were formed in the fusion zone depending on the amount of Al and Ti melted by the laser. These intermetallic phases are very brittle and the solid state cracks are formed near the Al side of the interface because of the stress developed after the solidification. In order to minimise the dissolution of Al in the fusion zone and to increase the toughness of the intermetallic phases, Nb foil is added as a buffer between the Ti alloy and Al alloy workpieces. It is observed that the partially melted Nb acts as a barrier to dissolve Al in the fusion zone and facilitates a good joining condition for welding of Ti alloy with Al alloy.

Liu *et al.*[269] developed a new kind of hybrid joining technique called 'laser (continuous) weld bonding' as an alternative to laser welding and adhesive bonding. With this method, a good joint between AZ31B Mg alloy and A6061 Al alloy was acquired.

Mai and Spowage[270] studied welding of steel to Kovar, copper to steel and copper to aluminium using a 350 W pulsed Nd:YAG laser. Laser welding of steel and Kovar resulted in significant intermixing of both materials within the fusion zone. Although the steel plate was completely melted, a complete metallurgical bond could not be obtained at the interface between the Cu plate and the weld metal, mainly due to the high thermal conductivity of copper. The weldability of copper to aluminium is regarded as relatively good. The Cu–Al phase diagram shows a wide range of Cu–Al phases that may be formed. In addition, non-equilibrium cooling conditions are known to promote the formation of a series of metastable phases. Under optimum processing parameters, crack free welds between copper and Al 4047 were obtained.

Kolodziejczak and Kalita[271] carried out keyhole LBW of dissimilar magnesium alloys AZ91 and AM50 with thicknesses of 4·5 mm using a CO_2 laser with a maximum power of 2·5 kW using helium as shielding gas. The microstructures, microhardness, tensile strength, bend strength and corrosion resistance of the welded zone were reported.

Qi and Song[272] reported on the role of Ni interlayer on the transformation of bonding mode of the joint between magnesium and mild steel developed by hybrid LATIG welding technique. The formation of Mg_2Ni intermetallic was reported at the interface between the magnesium alloy and nickel which was concluded to provide the metallurgical bonding of welding. On the other hand, the solid solution of Ni in Fe was detected along the side face of molten pool and accumulated at the bottom.

Liu and Zhao[273] reported the application of laser–gas tungsten arc hybrid welding technique for successful joining of AZ31B Mg alloy and 304 stainless steel. The mechanism of bonding was the chemical interaction between magnesium and iron. However, the formation of metallic oxide was responsible for the initiation of fracture during tensile testing.

Laser brazing

Like welding, laser is widely used in brazing non-ferrous metals/alloys and ceramics/glasses. Schubert *et al.*[274] have shown that a 1·4 kW diode laser (808 nm) with a rectangular spot can be very useful in cladding and brazing.

Biro *et al.*[172] have studied the effects of Ni and Au/Ni plating on laser brazing/welding of 200 μm thick aluminium, nickel, Kovar and cold rolled steel sheets in the lap joint configuration using a pulsed Nd:YAG laser and a range of weld process conditions. The strength of the joint was equivalent to that of the annealed base material. Significant gas porosity was observed at the interface between the Al weld pool and the unmelted Ni and Au/Ni plating layers. However, porosity did not affect the tensile shear strengths of the Al joints. Au/Ni braze increased the strength of the Au/Ni plated Ni specimens to that of the base material.

Laser soldering

Laser soldering of semiconductor devices needs extreme care and perfection. However, the precision of laser soldering is far superior to conventional soldering. For instance, 215 μm wide tape automated bonding leads could be soldered to a device with 430 μm centre to centre spacing by 98 ms diode pumped Nd:YAG laser at 5·8 W/lead laser power. As conventional soldering, one of the major concerns with laser soldering remains the mechanical property, particularly creep resistance at elevated temperatures. Guo *et al.*[275] have studied the creep behaviour in Cu and Ag particle reinforced composite and eutectic Sn–3·5Ag and Sn–4·0Ag–0·5Cu non-composite Pb free solder joints at 25, 65 and 105°C (representing homologous temperatures ranging from 0·61 to 0·78). Creep resistance in composite solder joints was significantly improved with Cu particle reinforcements, but not with Ag reinforcement. The strain at the onset of tertiary creep for Cu and Ag reinforced composite solder joints was typically lower compared with non-composite solder joints. The activation energies for creep were similar for all the solder materials. Brandner *et al.*[276] have explored soldering of thin or narrow Cu based electrical joints with solid state (1064 nm) and diode (808 nm) lasers to identify the mechanism of energy coupling, model the temperature rise and determine the process window. Laser soldering may be more beneficial for secondary fluxless reflow solder bumping than soldering in a furnace.

Berkowitz and Walvoord[277] have studied laser soldering of fine pitch printed wiring boards using a CW diode laser. The success of laser soldering of such semiconductor devices depends on the selection of laser, optimisation of laser parameters, surface preparation/cleaning, lead form configuration, solder selection and soldering process control.

Laser tissue joining

Laser has now opened new possibilities of joining organic substances including human and animal tissues and organs. Laser tissue welding uses laser energy to induce thermal changes in the molecular structure of the tissues, though low strength anastomoses and thermal damage of tissue are major sources of concern. On the other hand, laser tissue soldering is a bonding technique in which a protein solder is applied to the tissue surfaces to be joined, and laser energy is used to bond the solder to the tissue surfaces. The addition of protein solders to augment tissue repair procedures significantly reduces the problems of low strength and thermal damage associated with laser tissue welding. McNally *et al.*[278] have attempted to determine the optimal solder and laser parameters for tissue repair in terms of tensile strength, temperature rise and damage and the microscopic nature of the bonds formed. An *in vitro* study was performed using an 808 nm diode laser in conjunction with indocyanine green doped albumin protein solders to repair bovine aorta specimens. Liquid and solid protein solders prepared from 25 and 60% bovine serum albumins were compared. Increasing the laser irradiance or surface temperature resulted in an increased severity of histological injury. The strongest repairs were produced with an irradiance of 6·4 W cm^{-2} using a solid protein solder composed of 60% bovine serum albumin and 0·25 mg mL^{-1} indocyanine green with a surface temperature around 85$\pm$5°C and thermal gradient of $\sim$15°C across 150 μm thick solder strips. Post-soldering histological examination showed negligible evidence of collateral thermal damage to the underlying tissue. Microstructural study revealed albumin within the tissue collagen matrix and subsequent fusion with the collagen as the mechanism for laser tissue soldering. Thus, laser tissue soldering may be effective for producing repairs with improved tensile strength and minimal collateral thermal damage over conventional tissue welding. The tensile strengths of the welded shell membrane of hen eggs and the Wistar rat skin showed significant improvement using Forsterite and Cunyite tunable lasers with a range of wavelength from 1150 to 1500 nm.[279]

Summary and future scope

Laser joining is one of the earliest recorded applications of laser material processing, which is still considered a major field of useful application of laser. The principle reason for this popularity is the fact that heating by laser irradiation is possible for all kinds of engineering materials irrespective of the chemistry, state, bonding or size/geometry. This contactless direct heating is obviously a big advantage in joining a component with another of the same or different type by using laser just as a clean source of heating without the risk of any chemical reaction or change of the irradiated material. Obviously, joining special steels and alloys constitutes the main demand for laser welding or joining. Apart from routine joining, laser is useful in joining dissimilar materials such as steel to Al, steel to alumina, polymer to metal, etc. Even, joining of inorganic thin films, microelectronic devices, human/animal tissues and organic substances is now commonplace using laser. The main issues for future research concern joining materials with dissimilar physical (melting point, density and diffusivity) and mechanical (hardness and strength) properties and bonding (e.g. metallic to covalent). While several attempts have been made to develop mathematical models to predict the microstructure and width of the joint with similar materials, similar approaches now need to be extended for dissimilar materials. In this regard, capability to predict the microstructure, integrity and strength of the joints would be the core issue. The most useful contribution could be a multiscale model that can connect the atomic transport events to microstructural evolution of phases/phase aggregate and composition profile, and eventually to the mechanical properties in macroscopic length scale. In particular, investigations on the changes in microstructural and mechanical properties across the weldment as a function of the selected process/laser parameters, macroscopic length scales and material dimension/properties are needed. Perhaps, using filler rods can be dispensed with if mechanism for brittleness of the joints in certain combinations is well understood. On the other hand, using an intermediate barrier layer/metal that wets well with two dissimilar metals or components on either side but can prevent intermixing of these two components in the melt pool so that the formation of harmful brittle phases could be avoided would be a useful strategy. Apart from welding, attempts must be made to develop useful localised joining methods such as sintering and brazing to fabricate finished products of greater variety and challenges. Laser assisted welding could be extended to derive additional advantages of cladding, reclamation and refurbishing not only old and worn components but also to develop compositionally and functionally graded surfaces with tailored composition and microstructure. For further details about the technology and mechanism of laser joining, several recently published text books may be consulted.[2,4,166]

Laser machining

Laser machining refers to controlled removal of material at a given rate by laser induced concomitant melting and evaporation from the designated area/spot on the surface or bulk of the workpiece. If dimension or volume adds in laser forming or joining, laser machining leads to a reduction in size or dimension of a component. Laser machining can achieve almost all types of conventional machining operations such as drilling, cutting, cleaning, marking and scribing up to a limited dimension or depth.[1,8] Metals, ceramics, polymers and composites are all amenable to laser machining irrespective of the hardness and physical/chemical properties of the material. The entire process can be fully automated with high speed and precision retaining clean and sharp edges with minimum HAZ. Laser assisted cutting, drilling, cleaning (or paint stripping), marking and scribing can be applicable to any material including very soft polymers to extremely hard ceramics (through normal or oblique/inclined irradiation), ultrathin semiconductor chips to multilayer metallic films/sheets and simple/flat surfaces to complex/curved geometry. The discussion in this section will highlight the scope/versatility of laser machining and emphasise the recent advances and unresolved issues in using laser as a non-contact and non-contaminating tool for machining of engineering solids.

Laser cutting

Laser cutting does have a diversified application starting from thin sheet metal cutting for general purpose equipment (such as household appliances, electrical cabinets and automotive components) to thick section metal cutting (for trucks, buildings, stoves, construction equipment, shipbuilding, etc.). Research efforts that have been successfully conducted on laser assisted machining of engineering materials (both metallic and non-metallic) have been reported in the literatures.[280–323] Titanium alloys cut in an inert atmosphere are used in airframe manufacture. Aluminium alloys have similar advantages in cutting by using the laser. Cutting of radioactive material is another important area of application of laser cutting. Cutting of wood up to 1 in thickness may be useful in the die board industry, furniture industry, puzzle and gift industry, crafts and trophies, etc. Plastics, rubbers, composites, cloth, ceramics and such diverse materials can be cut by laser in textile industry. Hard and brittle ceramics such as SiN can be cut 10 times faster by laser than by diamond saw. In the following section, laser cutting of different materials is discussed in details.

Laser assisted cutting of ferrous metals and alloy

A simple empirical relation to describe the parameters for laser cutting of iron and steel with a high quality of kerf was established by Belic and Stanic.[280] This empirical relationship between laser power P (W), sheet thickness d (mm), cut width s (mm) and cutting speed V (mm min^{-1}) was established as: $P=390d^{0\cdot21}s^{\ 0\cdot01}V^{0\cdot16}$ (ignoring the variation in thermal properties of material with temperature). It agreed well with the experimental results obtained from laser cutting of iron, steel and stainless steel with a 500 W CW CO_2 laser. A systematic investigation of the effect of gas composition and pressure on CO_2 laser assisted cutting of mild steel was reported by Chen.[281] The gas mixtures used were oxygen, argon, nitrogen and helium. It was found that high purity oxygen was conductive for better cutting performance. Low pressure inert gas could not produce a good quality cut. The influence of high pressure oxygen jet on the cutting efficiency was studied by O'Neill and Gabzdyl[282] for thick sections of mild steel plates (20, 30 and 50 mm in thickness) with a CW CO_2 laser at an incident power of 1 kW. Striation formation at the cut edge is often encountered during oxygen cutting of mild steel, which was attributed to: cyclic variation in the driving force of the oxidation reaction owing to the changes in the oxygen partial pressure in the melt zone melt; and viscosity and surface tension effects associated with melt removal.[283]

Danishman *et al.*[284] measured the electron temperature and density of the surface plasma generated during CO_2 laser cutting of mild steel. Electron temperature and density were found to oscillate with cutting time. Electron number density decreased with increasing electron temperature. Electron number density also diminished when an argon+oxygen gas mixture was used rather than oxygen. It was concluded that the cut quality improved at high electron temperature and low electron number density.

Wang and Wong[285] carried out a detailed investigation on the machinability of galvabond (galvanised commercial steel sheet with spangled surface) steel sheet of 0·55, 0·8 and 1 mm in thickness using a CW CO_2

laser. A relation among the cutting speed, laser power and workpiece thickness was developed by statistical analysis for selecting the cutting parameters for process control and optimisation. Similarly, Prasad *et al.*[286] discussed the laser beam machining of metal coated steel sheet (mainly galvalume, galvanneal and galvabond) of 1 mm in thickness with a 500 W CW CO_2 laser and achieved an improved cut quality in terms of good surface finish, reduced kerf width and dross. An analytical model was developed to establish the parametric relation of the cutting process. The model suggested that an efficient choice of the process parameters was a prerequisite for minimum thermal damage of the coatings as the former had a significant influence on the topographical characteristics of the uncut through kerf, surface roughness and micro- and macromechanics of the cutting process. The comparison of magnetic properties of mechanically and laser cut uncoated non-oriented electrical steel suggests that the best magnetic property was obtained after mechanical cutting.[287] King and Powell[288] cut mild steel using continuous and pulsed CO_2 laser and concluded that pulsed laser irradiation is better than continuous one. Tabata *et al.*[289] carried out a detailed study of laser cutting of steel using CO_2, Nd:YAG and diode laser and compared the effect of different lasers on cutting quality. Zheng *et al.*[290] undertook laser cutting of stainless steel, aluminium and mild steel with CO_2 laser and showed that laser cuts had narrower kerf width with better accuracy than that obtained by mechanical cutting. Skvarenina and Shin[291] studied the machinability of compacted graphite iron by varying the depth of cut, feed and material removal temperature and then evaluating resultant cutting forces, specific cutting energy, surface roughness and tool wear. At a material removal temperature of 400°C, laser machining was successful to increase the tool life to 60% greater than conventional conditions at a feedrate of 0·150 mm rev^{-1} without affecting the microstructure. Surface roughness was improved by 5% as compared with conventional machining at a feedrate of 0·150 mm rev^{-1}. A 20% reduction in the cost of machining was also achieved by laser assisted machining.

Laser assisted cutting of non-ferrous metals

Among the common aeronautical materials, Al alloys are one of the most suitable materials for laser machining. Araujo *et al.*[292] did a detailed investigation of the microstructures generated from laser machining of 2024 Al alloy using CO_2 laser. Owing to the large α-liquid region at 4%Cu in the Al–Cu phase diagram and low laser absorption factor of solid Al (at 10·6 μm wavelength), the material partially melts before evaporation/sublimation of the liquid. On the other hand, the high pressure gas causes dripping of the high viscous melt and degrades the surface quality of the machined material. This promotes the generation of high stress concentration regions that might enhance its fatigue resistance. Indeed, the fatigue resistance of laser machined Al 2024-T3 aluminium alloy was found higher than that in the same alloy in as received condition.[293] Laser assisted cutting of Al based metal matrix composites was extensively studied to understand the microstructure, residual stress, mechanical properties (wear and fatigue) and process optimisation.[294–296] Yue and Lau[297] have studied the pulsed laser cutting of Al–Li/SiC metal matrix composite to examine the influences of laser cutting parameters on the quality of the machined surface. Proper process control seems effective to minimise the HAZ, improve the quality of the machined surface and predict the maximum depth of cut for the composite. The effects of femtosecond laser machining on surface characteristics and subsurface microstructure of a Nitinol alloy were studied by Huang *et al.*[298] The femtosecond laser machining produced an average surface roughness of 0·2 μm similar to the surfaces machined by precision milling.

Laser assisted machining also offers the ability to machine superalloys more efficiently and economically by providing the local heating of the workpiece before material removal by a single point cutting tool. The machinability of Inconel 718 under varying conditions was evaluated by examining tool wear, forces, surface roughness and specific cutting energy by Anderson *et al.*[299] With increasing material preheating temperature from room temperature to 620°C (by laser heating), the benefit of laser assisted machining was demonstrated by a 25% decrease in specific cutting energy, two- to threefold improvement in surface roughness and 200–300% increase in ceramic tool life over that in conventional machining. Moreover, an economic analysis shows significant benefits of laser assisted machining of Inconel 718 over conventional machining with carbide and ceramic inserts.

Laser assisted cutting of ceramics

The high hardness and brittleness of ceramics make them difficult and expensive to machine using conventional cutting methods or tools. Laser cutting provides a good means for cutting of ceramics in non-contact mode with excellent precision. In laser machining of ceramics, the machining process consists of three inherent stages, i.e. vapourisation cutting, melting and ejection and controlled fracture.[300] Lu *et al.*[301] studied the problem of cracking of ceramic plates during laser assisted cutting and identified that the most relevant parameters for laser cutting were laser power P and scan speed v, specific heat c, conductivity k and thermal expansion coefficient α. Tsai and Chen[302] attempted synchronous application of Nd:YAG and CO_2 lasers for laser cutting of thick ceramic substrates by controlled fracture. Tsai and Chen[303] proposed a new image processing system to determine the laser movement path by iterative path revision algorithm useful for monitoring laser assisted cutting.

Cutting of engineering ceramics such as silicon nitride (Si_3N_4) was studied by several groups including Hong *et al.*,[304] Hong and Li[305] and Rozzi *et al.*[306] with Q switched pulsed YAG laser. Considering the influence of the cut front shape on the absorption of the laser beam, a simplified two-dimensional mathematical model was developed based on the conservation of energy during pulsed laser vapourisation assisted cutting process. The experimental results showed that crack free cutting could be obtained by using high speed and multipass feed cutting routine. A detailed parametric investigation of laser assisted machining of aluminium nitride ceramic was undertaken with an UV and near infrared Nd:YAG laser by Gilbert *et al.*[307] The effect of pulse overlap and pulse frequency on material removal rate was studied. The effect of laser power, scan speed, mode of cutting and shield gas composition on cutting of thick ceramic tiles using a CW CO_2 laser was studied by Black and

Chua[308] and Black *et al.*[309] Crack formation was one of the major problems associated with the cutting process. Large thermal gradient between the cut zone and associated glazed zone was responsible for the crack formation. Proper choice of laser parameters was essential to allow sufficient time to cool and minimise the tendency for crack formation. In this regard, Lit and Sheng[310] opined that fracture initiation during laser cutting of alumina was primarily related to laser parameters.

Pascual-Cosp *et al.*[311] studied the cutting behaviour of highly vitrified ceramic materials and fine porcelain stoneware tile using Nd:YAG laser to determine the optimum processing zone for laser cutting. Preheating of the workpiece (to 1000°C) was beneficial to avoid catastrophic breakdown of the material under the action of laser. Besides cutting thick substrates, Park *et al.*[312] investigated the scope of precision micromachining of chemical vapour deposited diamond film using excimer laser and concluded that the principle and mechanism were identical to those for laser cutting of thick substrates.

Pfefferkorn *et al.*[313] studied the machining of magnesia doped partially stabilised zirconia using a CW CO_2 laser and concluded that the mechanism of material removal is a combination of brittle fracture and ductile deformation with the latter becoming dominant as the uncut chip thickness decreases near the penultimate point of tool contact.

Rebro *et al.*[314] showed the optimum processing conditions for a defect free laser assisted machining of pressure less sintered mullite ceramics using a CW CO_2 laser which otherwise is difficult due to its low thermal diffusivity and inferior tensile strength.

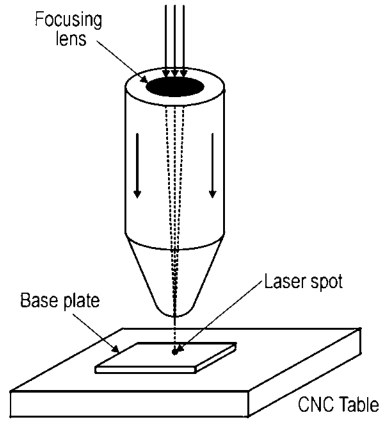
Laser assisted cutting of polymers

Laser cutting is equally useful in manufacturing special materials including circuit boards, resistance and functional trimming of circuits and microlithography. The growing use of the excimer laser in above operations may enable new and precision machining. Lum *et al.*[315] studied laser cutting of medium density fibreboard to determine the influence of laser parameters on the cut quality and optimum laser parameters (in terms of cutting speeds, material thickness and pulse ratio) for effective cutting of these fibreboards by CO_2 laser both under CW and pulsed modes of cutting.

Laser cutting of composites of aramid, graphite and glass cloth reinforced polyester were successfully implemented through studies on morphology of the cut surfaces by scanning electron microscopy and by numerical simulation of laser induced thermal changes.[316] The thermal properties of the fibres and matrix were identified as the principal factors that affected cutting performance. Laser machining has also been reported on polymer matrix composite.[317–323]

A theoretical model describing the temperature distribution during CO_2 laser cutting of thin thermoplastic material (used in the packaging industry) was developed by the Green function method.[324] It was shown that suitable control of laser beam diameter allowed two types of cutting operations: a simple cut, when the beam focus coincided with the surface and a cut with welding, when the beam was defocused.

Graff and Meijer[325] investigated CO_2 laser cutting of newly developed materials (synthetic aluminium laminates) for the aeronautic and automotive industry and found that these materials could be cut at the same speed as homogeneous aluminium alloys, although some damage on the synthetic layers was unavoidable. Based on these experimental results, a dedicated computer simulation model was developed using appropriate mass, power and force balance equations that considered splitting of the laminate in several horizontal layers according to the concerned thermophysical and optical properties.



10 Schematic of laser assisted drilling of thin sheets of metal, semiconductor or polymers with provisions for changing focus and rate of shroud gas flow

Laser drilling

Laser drilling is a machining operation by melting and evaporation of the workpiece using a high power laser beam. Metals/alloys, polymers and ceramics may be drilled by laser assisted drilling.[326–337] High power CO_2 laser or Nd:YAG laser is generally used to drill holes. Figure 10 shows the basic set-up used in laser drilling, which is not too different from that used for other laser machining processes except that the workpiece remains stationary with regard to the laser spot. Laser drilling can be done both in pulsed and CW modes with suitable laser parameters. The advantage of using laser is that it can drill holes not only vertically but also at an angle inclined to the surface (e.g. fine lock pinholes in Monel metal bolts). Mechanical drilling is slow and causes extrusions at both ends of the hole that have to be cleaned. Mechanical punching is fast but is limited to shallow holes with greater than 3 mm diameter. Electrochemical machining is quite a slow process though produces a neat and precise hole. Electrodischarge machining is expensive and slow with a typical rate such as 58 s/hole. Electron beam drilling is fast at 0·125 s/hole but needs a vacuum chamber and is more expensive than a Nd:YAG laser processing. In comparison, a Nd:YAG laser can produce a hole in 4 s to outsmart all other methods.[1]

Low *et al.*[326] studied the rate of sputter deposition concomitant with laser drilling in Nimonic 263 alloy for various laser processing parameters using a fibre optic delivered 400 W Nd:YAG laser. It was observed that shorter pulse widths, lower peak powers and higher pulse frequencies generated smaller sputter deposition areas.

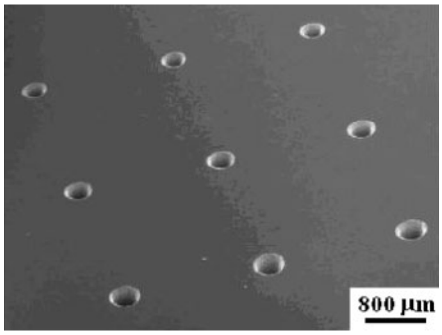
Laser micromachining

The ability to machine very small features such as holes into a metal, ceramic, semiconductor or polymer sheet/film by laser ablation with an unmatched precision, accuracy and speed has opened a very useful scope of application of laser material processing in microelectronic industry. For instance, holes with a diameter of 300 nm and a depth of 52 nm could be drilled in metal films with minimum distortion and HAZ using 200 fs and 800 nm pulses from a Ti:sapphire laser focused to a spot size of 3000 nm. Lehane and Kwok[328] have developed a novel method for improving the efficiency of laser drilling using two synchronised free running laser pulses from a tandem head Nd:YAG laser capable of drilling through 1/8 in thick stainless steel targets at a standoff distance of 1 m without gas assist. The combination of a high energy laser pulse for melting coupled with a properly tailored high intensity laser pulse for liquid expulsion results in efficient drilling of metal targets. The improvement in drilling is attributed to the recoil pressure generated by rapid evaporation of the molten material by the second laser pulse. Similarly, Zhu *et al.*[329] have carried out a detailed experimental study of drilling submicrometre holes in thin aluminium foils with thickness ranging from 1·5 to 50 μm, and W, Mo, Ti, Cu, Fe, Ag, Au and Pb foils of 25 μm in thicknesses with femtosecond Ti:sapphire laser pulses of 800 nm in width. The influence of laser parameters and material properties on hole drilling processes at submicrometre scale has been examined and a simple model to predict the ablation rate for a range of metals has been developed.

Laser precision machining has been applied to fabricate metallic photonic bandgap crystals operating in the microwave frequency.[338] Transmission measurements showed that the periodic crystals exhibited a cutoff frequency in the 8–18 GHz range allowing no propagation below this level. Furthermore, the cutoff frequency could easily be tuned by varying the interlayer distance or the filling fraction of the metal. The combination of plates with different hole diameters creates defect modes with relatively sharp and tunable peaks. Illyefaalvi-Vitez[339] has attempted utilisation of CO_2 and frequency multiplied Nd:YAG lasers (using five wavelengths, i.e. 10 600, 1064, 532, 355 and 266 nm) for drilling and direct patterning of copper clad glass fibre reinforced epoxy laminates, polyester foils and similar packaging structures. Laser processing was combined with through contacting of the generated vias by screen printing with polymer thick films, by wet chemical direct plating and by evaporation of thin metal layers. The results show promising opportunities for laser processing of metal layers and polymeric materials in microelectronic packaging.

Pronko *et al.*[340] demonstrated that for micromachining, femtosecond pulses offer an advantage over current technology of employing nanosecond pulses because of the reduced effects of thermal diffusion. Using the femtosecond pulses, holes with a diameter of 300 nm, roughly 10% of the spot size was produced, whereas the smallest diameter made using the nanosecond pulses was roughly 60% of the spot size. The depth of the femtosecond hole was 52 nm.

Different types of glass materials were machined by Nikumb *et al.*[341] using a short pulse solid state laser with pulse duration in the nanosecond to femtosecond range. The effect of pulse duration and other process parameters on the machined features was analysed to reveal the underlying thermal effects and non-linear processes. Edge quality, circularity, aspect ratio, formation of the redeposit material and machining rate were also studied



11 Array of through vias in 3C–SiC wafer with thickness of 400 μm drilled with solid state laser providing pulses with duration of 300 fs at wavelength of 1040 μm (Ref. 342)

with respect to the process variables such as focusing optics, laser power, wavelength and repetition rate. Arrays of drilled microhole patterns and fabricated microfeatures are demonstrated with discussions on their potential applications.

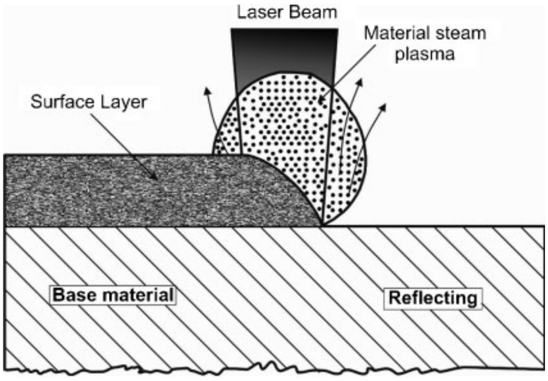
Zoppel *et al.*[342] studied laser ablation of silicon carbide with emphasis on the influence of pulse duration and laser wavelength on ablation efficiency. A KrF laser of 248 nm wavelength (34 ns pulse duration), ArF laser of 193 nm wavelength (25 ns pulse duration) as well as fs-UV experiments with a distributed feedback dye laser system amplified by an excimer laser, delivering 500 fs pulses at 248 nm were employed for laser processing. Additional experiments were performed with solid state lasers providing radiation at 1040 and 1026 nm at pulse durations of 300 and 400 fs respectively. Figure 11 shows an array of vias drilled in 3C–SiC at a wavelength of 1040 nm with a pulse duration of 300 fs. Despite the relatively low fluence of 10·6 J cm^{-2}, through via drilling was possible by employing trepan drilling, where the beam moves along a circular path.

Laser cleaning

Removal of small particles or continuous layers from a metal surface can be carried out by laser beam using a selected area irradiation at an optimum combination of incident power, interaction/pulse time and gas flowrate (that sweeps the dislodged atoms from the surface).[343–354] Figure 12 schematically shows the mechanism of laser cleaning. At the initial stage, a plasma plume is formed due to ionisation of the atoms vapourised from the surface and blocks the beam/surface contact. As the irradiation stops, the temporary compression on the surface changes into tension and causes spallation of the oxidised layer. A dramatic improvement in cleaning efficiency in terms of area and energy is possible when laser beam irradiates the workpiece at an oblique or glancing angle of incidence rather than direct or perpendicular irradiation of the surface.

The influence of pulsed laser irradiation on the oxide scale removal from the surface of stainless steels oxidised using a pulsed Nd:YAG laser was investigated by Psyllaki and Oltra.[343] At an appropriate angle of incidence, laser irradiation with an energy density of $\sim 10^4$ J m^{-2} resulted in expulsion of the oxide layer without any material removal from the underlying metal.

The side wall and bottom polymers resulting from reactive ion etching of via holes can be removed by a non-contact dry laser cleaning technique using pulsed



12 Schematic of laser assisted cleaning

excimer laser irradiation.[344] Similarly, laser cleaning is capable of removing the polymers by subthreshold ablation, even at fluences limited by the damage threshold (2500–2800 J m^{-2}) of the underlying Al–Cu metal film with titanium nitride (TiN) antireflective coating. Comparing ablation results obtained using Nd:YAG laser and excimer laser shows that although the shorter 7 ns Nd:YAG laser pulse gives a greater etch thickness than the 23 ns excimer laser pulse, it also tends to damage the metal films and the silicon substrates of the via wafers more easily.

Tsunemi *et al.*[345] have demonstrated that pulsed laser irradiation of oxidised metallic surfaces in an electrolytic cell under proper voltage conditions could be a promising new approach for effective removal of oxide films.

Schmidt *et al.*[346] studied the process characteristics for removing white chlorinated rubber coatings, pigmented with titanium white (TiO_2), from concrete surfaces utilising a 120 W CW diode laser. About 0·3 mm thick double coating layers were removed utilising oxygen process gas flow.

Laser removal of small copper particles from silicon wafer surfaces was carried out using appropriate laser radiation from near infrared (1064 nm) through visible (532 nm) to UV (266 nm) wavelength range.[347] It has been found that both 266 and 532 nm are successful in removing the particles from the surface, whereas 1064 nm was shown to be ineffective for removal of particles. The cleaning efficiency increased with a shorter wavelength. The mechanism of laser cleaning was explained by the evolution of tensile forces exceeding the adhesion force of the particle to the surface at given irradiation wavelengths.

Meja *et al.*[348] studied the cleaning of oxidised iron samples using a Nd:YAG laser to model the influence of wavelength in air and the role of environment on the cleaning behaviour. This study showed that cleaning occurs in a small energy density range irrespective of the wavelength of the laser. The adhesion force of the oxide layer influences the optimal laser parameters, and the repetition rate determines the minimal spot size for the desired cleaning. Furthermore, it was also concluded that the application of pulsed laser irradiation in liquid confinement under controlled electrochemical potential enhanced the removal efficiency of oxide films from metal surfaces.

The influence of excimer laser irradiation (of 193 nm in wavelength and 20 ns pulses and of 248 nm in wavelength and 20 ns and 500 fs pulses and fluencies between 8 and 19 J cm^{-2}) on the structure and morphology of plain and gold film coated mica surfaces was investigated.[349] Surfaces treated with laser fluences below the ablation threshold were well suited for controlled growth of metallic films, whereas above threshold treated samples form roughened surfaces with close packed arrays of cones. It was seen that the first step of the ablation process included reorientation of dipole domains on the surface and the ablated products possessed large kinetic energy and high directionality.

Single crystal SiC substrate was successfully photo etched at a remarkably high etch rate of 35 nm s^{-1} by (133, 141, 150, 160, 171 and 184–266 nm) multiwavelength laser ablation, combined with a chemical post-treatment in a solution of $HCl+H_2O_2+H_2O$ and $HF+H_2O$.[350] The analysis of the etched samples by scanning electron microscopy and scanning probe microscopy indicated that an array of square holes having well defined patterned structures and clean substrate surfaces were obtained. X-ray photoelectron spectroscopy analysis indicated that the SiC samples etched by multiwavelength laser have a similar stoichiometry after chemical post-treatment as the virgin SiC. The mechanism of high quality ablation using multiwavelength laser was discussed and compared with that for ablation using 266 nm single wavelength. The chemical post-treatment contributed to removing the residues from the laser photolysis of SiC.

Laser marking

Laser, as a source of heat, may be employed in marking of metallic, polymeric and glassy materials.[355–362] Laser ablation can be applied as a means for direct marking on plastics or polymer compounds as it is fast, economical and ecofriendly compared with the conventional marking processes.[355] Besides embossing a mark/design on a surface for identification or aesthetics, laser marking is also useful in non-contact and dry etching and printing of polypropylene or polyethylene plastics using a 532 nm laser pulse.[356] Similarly, laser irradiation can produce controlled microcracking or fracture of soda lime and borosilicate glasses.[357]

Beu-Zion *et al.*[358] have designed and constructed a CO_2 laser marking system based on a scanned flexible wave guide (silver halide infrared optical fibre or a hollow glass wave guide) capable of cutting, heating or marking on various surfaces. Recently, permanent markings of various colours on different ceramic and plastic substrates (with orasols dispersed palladium acetate organometallic films) have been made by a photo thermal deposition process using an argon ion laser (514 nm).[359]

Recently, attempts have been made to develop a dual laser writing scheme in which an unmodulated short wavelength red laser augments the writing process effectively by a longer- wavelength laser. Apart from increasing the thermal efficiency of the laser marking process itself, the dual beam writing scheme may decrease the mark width and the recorded mark length variability. Coupled with the increased resolution of the short wavelength read spot, these enhancements could improve the overall writing or marking performance.

Laser scribing

Scribing involves laser ablation of a groove or row of holes that form perforation lines to separate a large substrate into individual circuits. Laser scribing is extensively used for machining of ceramics in the microelectronics industry. The quality of scribing, particularly for silicon chips and alumina substrates is judged by the amount of debris produced and size of HAZ (the smaller the better for both). Lumpp and Allen[363] have drilled high aspect ratio, straight walled 60 and 300 μm diameter via holes in 635 μm thick aluminium nitride with or without a metallisation layer deposited on the via walls using an excimer laser. Attachment of a second substrate or metal sheet prevents damage to the back surface.

In fabrication of core laminations for motors and transformers, laser scribing may induce a reduction in hysteresis and eddy current losses. Laser scribing needs a very low power so as not to melt the surface. For example, pulsed excimer laser (248 nm and 23 ns) irradiation to scribe the grain oriented M-4 electrical steel has resulted in a 26% core loss reduction due to the beneficial thermal stresses that refined the magnetic domains and reduced the eddy current losses.[364] Similarly, 50% reduction in core loss in 50 μm thick amorphous alloy ribbon has been achieved by laser scribing at 2–5 mm line spacing at the same excitation power.[365] The mechanisms responsible for improvement include magnetic domain refinement, stress relaxation and inhibition of domain wall movement. Laser scribing also relieves the stresses that are induced in the material during manufacture. The scribe lines increase the surface resistivity of the material, resulting in reduced eddy current loss. Laser scribing of polycrystalline thin films used for solar cells can eliminate the frequently observed problem of ridge formation along the edges of scribe lines in the semiconductor films.[366]

Laser micromachining is a precision manufacturing technique, where pulses from ultrashort laser are used to induce micrometre sized structures on the surface or in the bulk of solid materials and an attractive technique for the high quality micromachining of many materials with minimum damage and maximum precision.[367,368] The technique has been established since over a decade with a potential scope of application in thin film deposition, microelectromechanical systems fabrication, surface micromachining, three-dimensional internal waveguide fabrication, nanotechnology and surgical applications.[369,370] A detailed basic understanding on micromachining with ultrashort laser pulses and technological development in this field for achieving an unprecedented level of accuracy at acceptable expenses has been presented by Dausinger *et al.*[371]

The mechanism of laser ablation is different for different materials. Hence, the optimisation of process parameters is essential for efficient and precision machining.[372] Cheng *et al.*[373] investigated the ultrafast laser microstructuring of Ti alloy, Al and Cu alloys using a Clark-MXR CPA2010 femtosecond laser system and a HighQ picosecond laser system. Ablation rate was found to increase with increasing repetition rate due to the combined effects of plasma, residual thermal energy and phase transition.

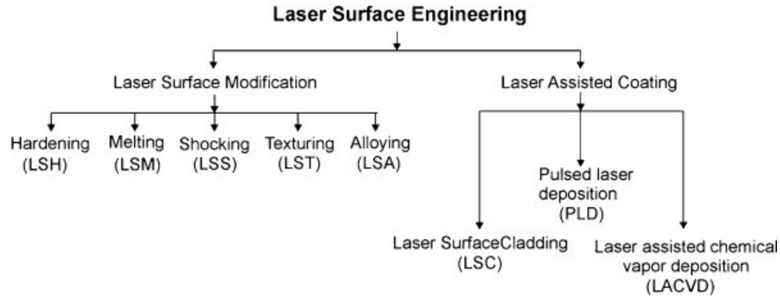
Summary and future scope

Laser machining requires controlled removal of material by vapourising within a narrow dimension incurring least heating and damaging of the surrounding region. The process can have many variations while removing controlled amount of materials: cutting, drilling, scribing, marking or cleaning. In all these processes, material removal without damaging the surrounding and maintaining precision and accuracy is a challenge. The precise spatial and temporal resolution of laser is certainly an advantage here. The recent advances are now based on selecting appropriate wavelength, using multiple beams, allowing inclined/oblique incidence and removing material in stages. Multiple beams can preheat, reduce thermal shock, remove debris or vapour and increase thickness of cut in sequence or stages. A large variety of materials starting from human/animal tissues to diamond can be laser machined. The mechanism of material removal obviously varies while slicing animal tissues or soft organics than when ultrahard diamond tools are reshaped or a silicon wafer is machined. However, an appropriate choice of laser power and wavelength is crucial for the success of the operation. Owing to the diversity of materials, geometry and conditions that can be handled, laser machining now often relies on more than one laser for the same operation. Use of advanced optics and better control of the sample stage are the other concerns with regard to improving the quality of the cut. Future challenges include increasing the capability to machine thicker sections, curved surfaces and dissimilar/heterogeneous materials. Similarly, assessment of material damage needs more close control and monitoring of the microstructural change/damage across the cut. Development of intelligent machines for machining diverse materials would require interfacing a vast database with the hardware. Both speed and precision in fully automated operation are the biggest advantage of laser machining. Thus, continued efforts are needed to model laser machining processes with suitable experimental validation of the predicted results.

Laser surface engineering

Failure of engineering components or materials due to chemical (corrosion and oxidation) or mechanical (wear and erosion) interaction is most likely to initiate from the surface because both external and internal surfaces/interfaces are more prone to environmental degradation, and intensity of externally applied load is usually the highest at the surface.[374] The engineering solution to minimise or eliminate such surface initiated failure lies in tailoring the surface composition and/or microstructure of the near surface region of a component without affecting the bulk.[2,3,374–376] In this regard, the more commonly practiced or conventional surface engineering techniques such as galvanising, diffusion coating, carburising and nitriding possess several limitations such as high time/energy/material consumption, poor precision and flexibility, lack in scope of automation/improvisation and requirement of complex heat treatment schedule. Furthermore, thermodynamic constraint of restricted solid solubility and kinetic limitation of thermally activated solute transport to solid state diffusion impose further limits to the level of improvement possible through these conventional or near equilibrium processes.[375–378]

Among the notable characteristic features, laser surface engineering enables delivery of a controlled quantum of energy (10^4–3×10^5 J m^{-2}) or power (10^8–10^{11} W m^{-2}) with precise temporal and spatial distribution either in short pulses (10^{-3}–10^{-12} s) or CW. The process is characterised by an extremely fast heating/



13 General classification of laser surface engineering

cooling rate (10^4–10^{11} K s^{-1}), very high thermal gradient (10^6–10^8 K m^{-1}) and ultrarapid resolidification velocity (1–30 m s^{-1}).[374–376] These extreme processing conditions often develop a novel microstructure and/or composition in the near surface region with large extension of solid solubility and formation of metastable including nanocrystalline and amorphous phases.

Figure 13 presents a brief classification of different laser surface engineering methods that involve mainly two types of processes. The first type is meant only for only microstructural modification of the surface without any change in composition (hardening, melting, remelting, shocking, texturing and annealing), while the second one involves both microstructural as well as compositional modification of the near surface region (alloying, cladding, etc.).

Laser surface engineering of ferrous alloys

As already mentioned, laser surface engineering is mostly utilised to enhance resistance to corrosion, oxidation, wear and similar surface degradation of engineering materials. In this section, we review the scope and current status of understanding about laser surface engineering of Fe based or ferrous alloys that involves surface hardening, melting, alloying or cladding using a high power laser beam to improve the surface dependent engineering properties of interest. Detailed studies under each category are discussed in the following subsections.[379–475]

Laser transformation hardening

Transformation hardening is a standard heat treatment for ferrous alloys (steel, cast iron, etc.) that involves heating to austenite (face centred cubic) phase and subsequent quenching to ambient or subambient temperature to enforce a shear induced austenite to martensite (body centred tetragonal) transformation instead of the equilibrium product/phase, i.e. ferrite (body centred cubic). In laser transformation hardening, only a thin surface layer of the substrate (instead of the entire bulk) is rapidly heated to the austenite phase field followed by rapid self-quenching (by conduction through the cooler bulk) to produce the desired martensitic microstructure.[379] Heating and cooling rates of 10^4 K s^{-1} or greater are typical of the laser surface hardening process and the entire thermal cycle in the irradiated volume may be accomplished in $<0{\cdot}1$ s. Kennedy *et al.*[380] have presented a detailed review on the fundamentals of laser hardening, associated benefits compared with the conventional hardening techniques and notable experimental studies particularly using high power diode lasers.

Prior microstructure of the substrate influences the final microstructure of the laser surface treated steel to a large extent. Grum and Slabe[381] observed the formation of nanometric precipitates with a diverse microstructure up to varying depth following laser surface hardening of maraging steel. Heidkamp *et al.*[382] studied the effect of laser beam hardening of TiN coated [by chemical vapour deposition (CVD)] AISI 4140 (EN42CrMo4), AISI A2 (ENX100CrMoV5-1) and AISI D2 (ENX153CrMoV12) substrates. A composite microstructure with strength and hardness gradients from surface to core together with a high surface hardness and wear resistance was achieved. Kulka and Pertek[383] heat treated the boride layers produced on 41Cr4 medium carbon steel (0·43%C and 1·02%Cr) using a CW CO_2 laser to obtain a laser surface modified surface with a reduction in hardness and brittleness. Similarly, laser heat treatment of carburised steel showed an improvement in microhardness and wear resistance.

Surface condition (roughness, residual stress and phase aggregate) is important in determining the characteristics and performance of the hardened layer. The effect of different types of absorption coating on laser surface hardening of C45E steel was studied by Grum and Kek[384] by using a photodiode with infrared radiation. The results confirmed that there was a strong correlation between the voltage signal of the infrared radiation and the size of the hardened layer on the steel surface depending on the type of absorbent chosen.

Heitkemper *et al.*[385] showed that fatigue and fracture behaviour of laser heat treated high nitrogen martensitic tool steel (X30CrMoN15) with a 3 kW Nd:YAG laser was significantly improved as compared with that of the untreated one. The residual stresses generated during laser treatment had a significant influence on crack initiation, while that produced during the transformation of retained austenite to martensite had a minor influence on mechanical behaviour. The impact resistance of laser surface hardened (using a 3 kW CW CO_2 laser) hypoeutectoid 2Cr13 martensite stainless steel was improved in laser treated sample as compared with that of the same processed under conventional heat treatment conditions.[386] The superior impact wear resistance of laser surface processed steel was attributed both to grain refinement and higher martensite content near the surface that resulted from rapid cooling. Iordanova *et al.*[387] carried out laser surface melting of 0·46 mm thick cold rolled low carbon steel by pulsed Nd:glass laser up to a thickness of 80 μm. Prior cold rolling developed residual tensile stress both along rolling and particularly along the transverse directions. Subsequent laser melting did affect the residual stress along the rolling direction, but significantly increased that along the transverse direction with a weaker and scattered texture.

Ashby and Easterling[388] developed a heat transfer model to predict the surface temperature during laser transformation hardening. The residual stress developed in the vicinity of the workpiece surface following repetitive pulse heating was predicted from the two-dimensional heat transfer and fluid flow model using a controlled volume approach by Shuja *et al.*[389] Laser annealing of aged 17-4 PH (AISI 630) steel (aged at 482°C) was carried out using a 5 kW Rofin–Sinar CO_2 laser.[390] Immersion test in the H_2S saturated solution indicated that laser annealed and H900 (peak aged) specimens had the similar ability to resist hydrogen induced cracking. Laser annealing led to an improved fatigue property predominantly because of the presence of residual compressive stress ahead of the laser annealed zone in the base metal. Amulevicius *et al.*[391] studied the influence of simultaneous laser irradiation and ultrasound on the phase evolution in the Fe–Si–C system (grey cast iron) by means of Mössbauer spectroscopy. The application of ultrasound reduced the amount of retained austenite in the samples by 2–3 times due to activated dissolution and redistribution of carbon between austenite and iron–carbon clusters. Chen and Shen[392] optimised the process parameters for laser transformation hardening of SNCM 439 steel using Taguchi methodology and fuzzy evaluation technique. Under optimal conditions, the hardness of laser hardened steel using a long pulsed Nd:YAG laser was improved from 52·5 to 63·9 HRC, the hardening width was increased from 0·43 to 0·89 mm and the erosion loss was reduced from 69·55 to 40·94 mg. Tobar *et al.*[393] optimised the process parameters for laser transformation hardening of a tool steel and simulated it experimentally. The method comprised an analytical three-dimensional transient model and a numerical finite element based approach, both concerned with solution of a transient heat conduction equation.

Molian[394] studied the effect of laser surface heat treatment on the microstructure of AISI 4340 steel and compared it with a conventional furnace heat treated one. The microstructure exhibited a mixture of dislocated and twinned martensites with an enhancement in volume fraction of retained austenite as compared with the conventional furnace heat treated (870°C treatment) specimens, though, prior austenite grain size and martensitic substructure remained essentially the same. Roy and Manna[395] have demonstrated that laser surface hardening, instead of laser surface alloying or melting, is more effective in enhancing hardness and wear resistance of unalloyed austempered ductile iron. The improvement in wear resistance following laser surface hardening is attributed to the martensitic surface with residual compressive stress. Lakhkar *et al.*[396] developed a numerical model to predict the back tempering in multitrack laser hardening of AISI 4140 steel. The model was capable of predicting the volume fraction of martensite that has been tempered and the corresponding hardness values. Temperature field in the laser hardening process was numerically simulated by Wang *et al.*[397] using a dedicate software to study the influence of energy density on laser hardening process. Simulation result was verified by a thermocouple temperature transducer measuring the specimen surface temperature following laser irradiation. Experimental curves of temperature versus time are in agreement with simulation results. The simulation results are useful in designing laser surface engineering experiments. Several other examples of laser surface hardening and melting of steel are listed in Table 3.[383–475]

Laser surface melting

Laser surface melting is another important and effective technique primarily to modify the microstructure of the near surface region. Lo *et al.*[398,399] showed that laser surface melting of 440C martensitic stainless steel was more effective in improving the corrosion resistance (in NaCl solution) by carbide refinement, while laser transformation hardening was useful in enhancing the cavitation erosion resistance by formation of an appropriate amount of martensite and austenite. Pan *et al.*[415] eliminated the carbides formed during sensitising treatment and homogenised the microstructure leading to a significant improvement in corrosion resistance by laser surface melting of austenitic stainless steel. Similar attempts to eliminate the susceptibility to intergranular corrosion of the cold worked and sensitised AISI 316 stainless steel and nitrogen bearing AISI 316L stainless steel were reported by Parvathavarthini *et al.*[402] and Mudali *et al.*[419] respectively. The improvement by laser surface melting is attributed to the complete dissolution of $M_{23}C_6$ type of carbides and suppression of reprecipitation due to rapid quenching. Laser surface melting of DIN X42Cr13 samples showed a higher abrasive wear resistance than the same after tempering treatments due to strain induced transformation of austenite into martensite and the work hardening of austenite during the wear tests.[411] Wilde *et al.*[427] showed that laser surface melting of AISI 4135 low alloy steel resulted in reduction in the hydrogen absorption kinetics as measured by the permeation test, and an improved resistance to hydrogen induced fracture in the slow strain rate tensile test under galvanostatic charging conditions.

Laser glazing is a process of rapid melting and solidification of materials using a high power laser beam as a source of heat.[444] Laser glazing was successfully applied to improve the surface properties of the VPS NARloy-Z coatings developed on steel.[445]

Laser shock processing

Laser shock processing involves rapid irradiation of the component surface with a laser (at a power density of $\sim 10^{12}$ W m^{-2}) resulting in generation of shock wave (due to a volume expansion of the plasma plume formed on the surface) and subsequent alteration of microstructure/state of stress. This processing has the capability to improve material hardness and fatigue strength in a number of alloys.[413] The effects of laser shock processing (with a Nd:glass phosphate laser with a 600 ps pulse width and an up to 120 J pulse energy at power densities $>10^{16}$ W m^{-2}) on the microstructure, microhardness and residual stress of low carbon steel were studied by Chu *et al.*[413] Surface hardness increased up to 80% after the laser shock processing up to 100 μm in depth and introduced residual compressive stress on the surface. The strengthening effect was attributed to the presence of a high dislocation density. Peyre *et al.*[408] showed that laser shock processing offered a better pitting corrosion behaviour of AISI 316L stainless steel than that obtained by shot peening in saline environment.

Dumitru *et al.*[406] used a Q switched Nd:YAG laser to develop microholed microstructure on AISI 440C

Table 3 Summary of selected studies on laser surface hardening and melting of steel

System	Results	Reference	Year
AISI 304 stainless steel	The fatigue behaviour and stability of the residual stress state and new surface microstructure of laser shock peened and deep rolled austenitic stainless steel AISI 304 was investigated	Nikitin and Altenberger[412]	2007
General	Review on the application of high power diode laser in surface hardening	Kennedy *et al.*[380]	2004
AISI H13 steel plate	The method comprises an analytical three-dimensional transient model and a numerical finite element method one, based both on the solution of a transient heat conduction equation	Tobar *et al.*[393]	2006
Carburised and borocarburised 15CrNi6 low carbon steel	Laser heat treatment improves the hardness and wear resistance of carburised and borocarburised steel	Kulka and Pertek[383]	2007
High nitrogen martensitic steel	Laser heat treatment improved the tribological and fatigue properties	Heitkemper *et al.*[385]	2003
2Cr13 steel	Impact property was improved by CW CO_2 laser treatment attributed to grain refinement and higher martensite content near the surface	Shao *et al.*[386]	2003
AISI 440C martensitic steel	Cavitation erosion resistance was significantly increased by laser transformation hardening	Lo *et al.*[398,399]	2003
Nodular cast iron	Nodular cast iron was modified by laser microprecision treatment. The substantial increase in the wear resistance was found	Xia *et al.*[400]	2002
Low C steel (0·05%C steel)	Pulsed Nd:YAG laser treatment caused scattering of crystallographic texture and introduction of significant tensile residual macrostresses transverse to the rolling direction	Iordanova *et al.*[387]	2002
Steel	The effect of repetitive pulse heating on the residual stress developed and its distribution was calculated using a mathematical model	Shuja *et al.*[389]	2002
Stainless steel	A combination of analytical and numerical model to modify and predict the process parameters for laser surface hardening of stainless steel	Yanez *et al.*[401]	2002
AISI 316 stainless steel	Laser surface melting is effective in desensitisation	Parvathavarthini *et al.*[402]	2001
AISI 304 stainless steel	Tendency for pitting decreases due to dissolution of nitrogen and formation of nitrides	Conde *et al.*[403]	2001
Austempered ductile iron	Laser surface hardening is superior to laser surface melting in terms of microhardness, residual stress and adhesive wear resistance	Roy and Manna[395]	2001
Austenitic 316 stainless steel and high speed tool steel	Morphology of surface microstructure and topographical features were studied as a function of laser parameters	Trtica *et al.*[404]	2001
17-4 PH steel	Stress corrosion cracking and fatigue crack growth behaviour were determined in 17-4 PH stainless steel by laser surface annealing treatments	Tsay *et al.*[390]	2001
Borided steel	Laser surface treatment of borided steels results in the deterioration of the case microstructure and properties	Gopalakrishnan *et al.*[405]	2001
AISI 440C steel	Microstructural modification on the steel was done by a Q switched Nd:YAG laser. Tribological test showed a significant improvement in wear behaviour	Dumitru *et al.*[406]	2000
Fe–Si–C system (grey cast iron)	Simultaneous laser irradiation and ultrasound had a great influence on dissolving carbon and disintegrating oversaturated γ-Fe(C) solution	Amulevicius *et al.*[407]	2000
AISI 316L stainless steel	Laser peening (LP) improved the pitting corrosion r esistance of AISI 316L stainless steel	Peyre *et al.*[408]	2000
SNCM 439 steel	Optimum process parameters for laser surface hardening were derived by Taguchi methodology and fuzzy evaluation approach	Chen and Shen[409]	1999
Mild steel (EN8)	Interaction of CO_2, Nd:YAG, excimer and high power diode lasers with surface of mild steel (EN8) changed the wettability characteristics	Lawrence and Li[410]	1999
DIN X42Cr13 tool steel (Fe–0·4 wt-%C–13 wt-%Cr)	Abrasive wear resistance of laser surface melted sample decreased with increasing volume fraction of retained austenite	Colacào *et al.*[411]	1999
Low C steel	Laser shock induced plastic deformation resulted in extensive formation of dislocations with increased surface hardness	Chu *et al.*[413]	1999
Carbon steel	Conversion electron Mossbauer spectroscopy study was conducted to observe the effect of CO_2 laser surface irradiation on carbon steel	Agudelo *et al.*[414]	1999
321 austenitic stainless steel	Laser surface melting eliminated the carbide formed by sensitisation and homogenised the microstructure leading to improvement of localised corrosion resistance	Pan *et al.*[415]	1998

Table 3 Continued

System	Results	Reference	Year
AISI 4150 steel	Resistance to fatigue crack propagation rate is significantly improved in laser surface heated specimen as compared with that of as received AISI 4150 steel (0·47C, 0·16Ni, 1·04Cr, 0·21Mo, 0·25Si, 0·80Mn, 0·015P and 0·02S)	Tsay and Lin[416]	1998
AISI 420 stainless steel	Amount of retained austenite in AISI 420 stainless steel is r educed by reducing laser scanning speed and increasing the laser power density	Colacào and Vilar[417]	1998
DIN X42Cr13 martensitic tool steel	Laser melted steel was secondary hardened at 600°C which is 100°C higher than conventionally heat treated steel (composition: C: 0·47, Cr: 12·82, Mo: 26, Ni: 0·19, Si: 0·40, Mn: 0·78, P: 0·024 and S: 0·011)	Colacào and Vilar[418]	1998
Nitrogen bearing 316L stainless steel	Laser surface melting is helpful in desensitisation of stainless steel by complete dissolution of $M_{23}C_6$ type of carbides and suppression of reprecipitation due to rapid quenching	Mudali *et al.*[419]	1998
Pearlitic rail steel	Laser surface melting produces a thin glazed layer with better hardness and lower friction rate	Di-Melfi *et al.*[420]	1998
EN31 bearing steel	Single or double glazed laser surface hardening tracks have better wear and friction property	Zhang *et al.*[421]	1997
Pure iron and low alloy steel	The effect of prior austenite grain size on the hardness of the hardened layer is established	Das[422]	1997
1Cr17Ni2 stainless steel	A detailed microstructural investigation of laser melted steel was conducted	Song and Shen[423]	1997
304L stainless steel	Laser surface melting of 304L stainless steel (using a CO_2 laser)	Akgun *et al.*[424]	1995
Ductile cast iron	Laser surface melting led to sevenfold enhancement in cavitation erosion resistance of ductile iron	Gadag and Srinivasan[425]	1995
Ductile iron	The novel Fe based material containing rapidly solidified eutectic cementite and the ultrafine graphite particles was fabricated by laser melting of ductile iron	Wang and Bergmann[426]	1995
AISI 4135 low alloy steel	The effect of laser heat treatment of steel on stress corrosion and hydrogen embrittlement was studied	Wilde *et al.*[427]	1995
Ductile iron	Laser surface melting led to a significant improvement in wear resistance attributed to the ultrafine microstructure and high hardness	Gadag and Srinivasan[428]	1994
2Cr13 stainless steel	Effect of laser surface treatment on the fretting wear resistance of 2Crl3 stainless steel was studied	Yang *et al.*[429]	1994
Fe–26Mn–7Al–0·9C	A microstructural study and aging behaviour of laser surface melted austenitic steel was undertaken	Tjong and Tsang[430]	1993
Fe–Mn–Al austenitic steel	Laser surface treatment results in significant improvement in fatigue crack growth resistance of a Fe–Mn–Al austenitic steel	Saxena *et al.*[431]	1993
0·4%C steel	Laser surface melting of 0·4%C led to a marginal improvement in hardness without any change in erosion behaviour	Rao *et al.*[432]	1993
304L stainless steel	Laser surface melting decreased the tensile strength and hardness, which was attributed to an overall decrease in dislocation density	Akgun *et al.*[433]	1992
AISI 420 steel	Laser surface melting was proved to be beneficial to i mprove the corrosion resistance	Escudero *et al.*[434]	1992
EN40-B substrate	The metallurgical and mechanical properties of laser nitrided surface was reported	Karamis and Yilbas[435]	1991
16 wt-%Cr white cast iron	Laser surface melting and hardening reduced the erosion rate in distilled water considerably	Tomilson and Talks[436]	1990
Pearlitic grey and ductile cast iron	The erosion rate of laser treated grey and ductile iron were 5 and 25 times enhanced	Molian and Baldwin[437]	1987
Pure iron and plain C steel with up to 0·2%C	Laser surface melting and resolidification developed a hard homogeneous martensitic structure	Hegge and de Hosson[438]	1987
AISI 4340 steel	The effect of laser heat treatment on the microstructures of the hardened layer was studied and compared with the same by conventional processing	Molian[394]	1981

stainless steel for improving the friction and wear properties. The hole diameters of about 5–10 μm and structures with spatial frequencies >60 mm^{-1} were achieved. The depth of the laser induced microholes was up to 5–8 μm (one pulse per hole) and could be increased if more pulses were applied for each microhole. Tribological tests, performed on samples structured with arrays of microholes having a diameter of 10 μm and different spatial frequencies (from 30 to 60 mm^{-1}) showed significant increase in the lifetime of the laser processed samples, in comparison with that in unstructured ones.

Laser surface alloying

Among the various laser surface engineering methods, laser surface alloying involves melting of predeposited or

plasma deposited layer. Standard immersion test was conducted in a 3·56 wt-%NaCl solution to compare the effect of laser surface alloying on the pitting corrosion resistance. A detailed study of the kinetics of erosion corrosion (in terms of weight loss per unit area followed by continuous circulation of the samples at 750 rev min^{-1} for 10–75 h in a medium containing 20 wt-% sand in 3·56 wt-%NaCl solution) in the stainless steel substrate before and after laser surface alloying (with 1210 MW m^{-2} power density and 31·7 mm s^{-1} scan speed for deposit thickness of 250 μm) showed a significant decrease in the kinetics of erosive corrosion loss due to laser surface alloying.[458] It was thus concluded that laser surface alloying was capable of imparting an excellent superficial microhardness and resistance to corrosion and erosion–corrosion properties to austenitic stainless steel due to Mo both in solid solution and as precipitates.

Oxidation is another serious mode of surface degradation that gets aggravated under unabated counter ionic transport of cations and anions at elevated temperature. Unlike electrochemical corrosion, oxidation occurs in dry condition (without aqueous medium) and solid state ionic transport through the oxide scale. In the past, several attempts have been made to enhance the resistance to oxidation by laser surface alloying, cladding and similar laser surface engineering techniques.[475,483,489,461] Pillai *et al.*[447] have obtained enhanced oxidation resistance (at 873 K for up to 200 h) of plain carbon steel following laser surface alloying with Al due to several intermetallic phases such as $Al_{13}Fe_4$ and Al_2Fe_2. Thermal barrier coatings could also be effective for improving oxidation resistance of stainless steel.[485] However, such coatings are brittle and hence are applied with a favourable composition gradient such as applying a bond coat (Ni–22Cr–14Al–1Y) between the topcoat (ZrO_2+7·5 wt-%Y_2O_3) and stainless steel substrate before single or multipass laser melting treatments. Oxidation tests at 1200°C indicated that double pass laser surface melting was more effective to impart a better oxidation resistance.[485] Similar attempts of improving oxidation resistance have been made with Incoloy 800H by laser surface alloying with Al[461,489] and by laser surface cladding of AISI 316L stainless steel with Fe–Cr–Al–Y alloy coatings.[489] In all these cases, it appears that the degree of enhancement in oxidation resistance by laser surface engineering primarily depends on stability, adherence, imperviousness and strength of the complex oxide scale comprising multiple oxide on top of the base metal.

The effect of laser melting on the oxidation resistance of plasma spray deposited stainless steel on mild steel substrates was studied by Khanna *et al.*[491] The oxidation resistance was improved significantly by plasma spray deposition which was reported to improve further improved by laser melting.[491] The adherence of the scale, tested using an acoustic emission technique and thermal cycling, showed significant improvement. The comparative tribological behaviour of plasma sprayed and laser alloyed coatings was also studied. Wear resistance of laser melted surface followed by plasma spraying was improved as compared with plasma sprayed and as received substrate.[492]

Manna *et al.*[450] attempted enhancing the high temperature oxidation resistance (>873 K) of 2·25Cr–1Mo ferritic steel by laser surface alloying with codeposited Cr using a 6 kW CW CO_2 laser. The main process variables chosen for optimising the laser surface alloying routine were laser power (from 2 to 4 kW), scan speed of the sample stage (150–400 mm min^{-1}) and powder feedrate (16–20 mg s^{-1}). Isothermal oxidation studies in air at 973 and 1073 K for up to 150 h revealed that laser surface alloying had significantly enhanced the oxidation resistance attributed to the presence of an adherent and continuous Cr_2O_3 layer.[450] Metal matrix composites generally possess an enhanced wear resistance than that of the base substrate or the matrix. Development of composite layer on the surface by conventional means is extremely difficult. Laser melting of substrate and subsequent feeding of ceramic particles into the molten matrix is an effective means of developing composite layer on the surface through a process called laser composite surfacing.

Several attempts have been made to develop a composite layer on metallic matrix by this technique. Cheng *et al.*[446] have added a mixture of WC–Cr_3C_2–SiC–TiC–CrB_2 and Cr_2O_3 to produce a metal matrix composite surface on stainless steel UNS S31603. Following laser surface melting, cavitation erosion resistance improved in all cases except for Cr_2O_3. Wu and Hong[454] have attributed the improvement in hardness and corrosion/wear resistance of stainless steel to the presence of amorphous phase in the Zr rich surface (with 7·8–14·5 at-%Zr) following laser surface alloying of stainless steel with Zr nanoparticles. Agarwal and Dahotre[455] have reported a substantial improvement in resistance to adhesive/abrasive wear of steel following laser surface alloying with TiB_2. The surface composite layer containing ~69 vol.-%TiB_2 particles recorded an elastic modulus of 477·3 GPa. Similar improvement in wear resistance of mild steel was reported by Tondu *et al.*[457] due to the formation of an FeCr+TiC composite coating formed by laser assisted self-propagating high temperature synthesis. Earlier, Zhukov *et al.*[490] achieved a similar carbide dispersed composite coating on steel by laser assisted self-propagating high temperature synthesis.

Pelletier *et al.*[483] applied an Fe–Mn–C Hadfield steel coating by laser surface cladding on steel and achieved significant improvement in hardness (>800 VHN), elastic modulus (210 GPa) and yield strength (1200 MPa) mostly due to the deformation induced twinning transformation. Laser surface alloying of A7 tool steel with Ti was reported to reduce the dry sliding friction coefficient due to the formation of a low friction transfer film.[462] Similarly, stellite coating by laser surface cladding is a standard method of reducing wear loss of ferrous substrate.[488] Besides providing with a harder and tougher surface, stellite coating develops a tougher oxide scale and reduces wear under severe condition. Improvement in hardness and wear resistance has been achieved by laser surface cladding of martensitic steel with Ni alloy[484] and laser surface alloying of ferritic steel with Cr_3C_2+SiC.[467]

Recently, an attempt has been made to develop a hard *in situ* boride dispersed composite layer on the surface of AISI 304 stainless steel substrate to improve its wear resistance property.[478] Laser processing was carried out by melting the surface of sand blasted AISI 304 stainless steel substrate using a CW CO_2 laser and simultaneous deposition of a mixture of K_2TiF_6 (potassium titanium hexafluoride) and KBF_6 (potassium hexafloroborate) (in the weight ratio of 2 : 1) using Ar as shrouding environment. Powder feedrate was maintained constant at 4 g min^{-1}. Irradiation results in dissociation of a predeposited mixture along with a part of the stainless steel substrate, intermixing and rapid solidification to form the composite layer on the surface. The microstructure of composite layer consisted of dispersion of titanium boride particles in AISI 304 stainless steel matrix with an uniform volume fraction of particles and an improved microhardness (250–350 VHN as compared with 220 VHN of the AISI 304 stainless steel substrate) and wear resistance.[578] The mechanism of improvement in microhardness in the composite layer is due to both grain refinement and dispersion strengthening. It may be noted that both laser power and scan speed should be carefully chosen in order to achieve an improved microhardness, microstructural homogeneity and uniformity of dispersion in the composite layer. Several other examples of laser surface alloying and cladding of steels are enlisted in Table 4.

Laser surface engineering of non-ferrous alloys

Surface dependent degradation by wear, oxidation and corrosion is as much a problem in non-ferrous metals and alloys as that in ferrous alloys. Thus, laser surface engineering is widely applied to Al, Ti, Cu, Mg and other important non-ferrous metals and alloys to extend the service life of components. In this section, the scope and current status of understanding regarding the application of laser assisted surface engineering to enhance surface dependent properties of non-ferrous alloys has been briefly reviewed. For a ready reference, Table 5 summarises the major work done in the recent past in this area.[445–585]

Laser surface engineering of Al alloys

Several attempts have been made in the past to improve corrosion resistance of Al based alloys by laser surface melting, alloying, cladding or remelting. Table 5a summarises the notable outcome on laser assisted surface engineering of Al based alloys.[493–556] Grain refinement and microstructural homogeneity are responsible for better corrosion resistance in laser surface melted alloy systems with dispersion of intermetallic and/or amorphous phases.

Yue *et al.*[512] surface melted 7075-T651 Al alloy using an excimer laser with pulse energy of 800 J mm^{-2}. Within the resolidified laser melted layer, the grain boundaries were found to be free from precipitates. The results of the stress corrosion cracking tests showed that the untreated specimens after the 30 days immersion test was severely attacked with intergranular cracks that penetrated deep inside the specimen. In contrast, no intergranular cracks were found in the laser treated specimen. The electrochemical impedance measurements conducted during the stress corrosion cracking test showed that the film resistance of the laser treated specimen was always higher than that of the untreated specimen.

Chan *et al.*[493] modified the surface microstructure of 7075-T651 aluminium alloys using an excimer laser (KrF) with a wavelength of 248 nm. A detailed transmission electron microscopy study revealed that laser surface melting caused reductions both in number and size of constituent particles and refinement of the grain size. As a result, the corrosion resistance of the aluminium alloy was improved. Under dry fatigue condition, the total fatigue life of the laser treated specimens, in which the crack initiation period is of considerable significance, was lower than that of the untreated specimens. However, after shot peening, the fatigue life of the laser treated specimens was recovered. The fatigue resistance of the shot peened laser treated specimens, tested in 3·5 wt-%NaCl solution with 48 h prior immersion, was greater than the untreated specimens with an increase of two orders of magnitude in fatigue life. This was primarily due to the elimination of surface defects and the reduction in corrosion pits. Li *et al.*[513] surface melted 2024-T351 aluminium alloy using a CW CO_2 laser. The laser treatment changed both the anodic polarisation behaviour and the form of localised corrosion in deaerated 3%NaCl solution. Immersion tests in the same solution under the condition of natural aeration showed that both intergranular and pitting corrosion occurred with pits distributed mainly along the rolling direction in the as received alloy while, only pitting corrosion was present with pits distributed uniformly for the laser surface melted material. This difference in corrosion behaviour as a result of laser surface melting is attributed to changes in the uniformity of distribution and composition of the second phase particles present in the alloy.

Badekas *et al.*[496] irradiated an Al–4·5%Cu alloy with an excimer laser. The residual stress of the treated surface changed from tensile to compressive with an improved microhardness. Gremaud *et al.*[510] laser surface melted Al–Fe alloy (with Fe content varying from 0·25 to 8 wt-%) using a 1·5 kW CW CO_2 laser and observed significant influence of process parameters on the morphology and degree of fineness of the precipitates/grains.

Hari Prasad and Balasubramaniam[495] laser irradiated an Al–Li–Cu alloy (of nominal composition 2·5%Li, 1%Cu and 0·12%Zr) with an Nd:YAG laser for the purpose of surface modification and studied its oxidation behaviour at 450°C in air by thermogravimetry and compared with the untreated specimen. Parabolic oxidation kinetics was studied for both the laser treated and untreated specimens. The laser treated specimen exhibited improved oxidation resistance. X-ray diffraction (XRD) analysis indicated that the major component of the scale was Li_2CO_3. The improved oxidation behaviour of the laser treated specimen was related to the compact oxide structure produced on its surface during oxidation. The nature of scale formation on the surface was also addressed. Wang *et al.*[62] melted a series of Al–Si alloys with various Si contents using a 2 kW CW CO_2 laser. A desirable kind of microstructure in laser melted zone of Al–Si alloys was obtained within the power density range of 156–208 W mm^{-2}. The degree of hardening of the laser melted surface increased with silicon content. It was found that laser treatment refined the grains and altered the morphology of eutectic silicon from lath-like to coralline-like with a higher Si content in Al matrix as compared with equilibrium Si content in Al. X-ray diffraction analysis revealed that the lattice parameter of Al reduced after laser melting. Wong *et al.*[511] reported that the corrosion behaviour of laser surface melted Al–Si alloy (using a CW CO_2 laser) was significantly improved in 10%H_2SO_4 and 10%HNO_3 solutions, although similar level of improvement was not observed in 10%HCl + 5%NaCl solution.

Table 4 Summary of selected studies on laser surface engineering of ferrous metals and alloys

a. Laser surface alloying

System	Results	Reference	Year
31603 stainless steel (CrB_2, Cr_3C_2, SiC, TiC, WC and Cr_2O_3)	Erosion resistance improves considerably for all carbides and borides except Cr_2O_3	Cheng *et al.*[446]	2001
Steel (Al)	Laser surface alloying with Al forms several aluminides that impart good oxidation resistance at 600°C up to 200 h	Pillai *et al.*[447]	2001
Electroless nickel coated mild steel	Laser melting exhibited a better corrosion resistance in deaerated Na_2SO_4 solution	Renaud *et al.*[448]	1990
AISI 1040 and AISI 316L	Following laser surface alloying NiCoCrB with corrosion resistance is improved in mild steel but deteriorated in stainless steel	Kwok *et al.*[449]	2001
Cr–Mo steel (Cr)	Oxidation resistance improved at 800–1000°C due to Cr_2O_3 rich scale	Manna *et al.*[450]	2000
UNS S31603 stainless steel	Laser surface alloying with Co, Ni, Mn, C, Cr, Mo, Si, SiFe, Si_3N_4 and NiCrSiB was carried out. A significant i mprovement in corrosion resistance was achieved with Si and Si_3N_4	Kwok *et al.*[451]	2000
UNS S31603 austenitic stainless steel (Co, Ni, Mn, Cr and Mo)	Resistance to erosion is improved due to dispersed ceramic/intermetallic phases but pitting resistance is decreased	Kwok *et al.*[452]	2000
AISI 304 stainless steel (Si)	Si turns the matrix ferritic. Post-laser surface alloying homogenisation improved corrosion resistance significantly	Isshiki *et al.*[453]	2000
Austenitic stainless steel (nanoZr)	Hardness and wear/erosion resistance improves significantly due to dispersion of Zr rich amorphous phase	Wu and Hong[454]	2000
AISI 1040 (TiB_2)	Laser surface alloying improves resistance to adhesive/ abrasive wear and reduces friction	Agarwal and Dahotre[455]	2000
Plain carbon steel (TiB_2)	Complex oxide layers develop on steel surface exposed to 600–1000°C following parabolic growth rate	Agarwal *et al.*[456]	2000
Steel (FeCr–TiC)	Laser surface remelting homogenises the microstructure and improves wear resistance	Tondu *et al.*[457]	2000
AISI 304 stainless steel (Mo)	Significant improvement in microhardness, pitting and erosion corrosion resistance by laser surface alloying of AISI 304 stainless steel with Mo	Dutta Majumdar and Manna[458]	1999
Fe (Ni)	A theoretical model was proposed to calculate the mixture ratio between the alloy and substrate D_j and the transport coefficient m_i of the i_{th} element from the added material and the substrate	Xie[459]	1999
Cast iron (Ni)	Pulsed excimer laser irradiation of nickel coated cast iron improved the microhardness and corrosion resistance	Panagopoulos *et al.*[460]	1998
Inconel 800H (Al)	Laser surface alloying with Al develops an Al rich surface that considerably improves resistance to oxidation up to 1000°C	Gutierrez and Damborenea[461]	1997
A7 tool steel	Excimer laser surface treatment and surface alloying with Ti had no discernible effect on mechanical properties but the friction coefficient was significantly reduced	Jervis *et al.*[462]	1997
304 stainless steel and 2·25Cr–1Mo steel	Laser surface melting improved the tribological properties of plasma sprayed coated 304 stainless steel and 2·25Cr–1Mo steels with WC, Co, Mo and Cr	Khedkar *et al.*[463]	1997
Mild steel (Ni–P)	Laser surface alloying of Ni–P on mild steel deteriorated the corrosion resistance because of the presence of Fe in the alloyed layer	Garcia-Alson *et al.*[464]	1996
316L and P900N stainless steel	Laser surface melting of Fe–Cr–Ni Si_3N_4 and Fe–Cr–Mn–Si_3N_4 powders introduced nitrogen and exhibited higher resistance to pitting corrosion	Huang *et al.*[465]	1996
Mild steel (0·21C–1·46Cr–3·52Ni–0·80W)	Laser surface cobaltising of mild steel achieved a Co content of 9·96 wt-%. The performance of the surface at elevated temperature was improved	Ding *et al.*[466]	1996
35 NCD 16 ferritic steel (Cr_3C_2/SiC)	Hardness and oxidation resistance were increased due to Si containing oxides	Gemelli *et al.*[467]	1996
AISI 304 stainless steel (Mo and Ta)	Laser surface melting produces δ-ferrite and deteriorates corrosion. Laser surface alloying with Mo is useful	Akgun and Inal[468]	1995
Mild steel (Fe, Cr, Si and N)	Laser surface alloying with Fe–Cr–Si–N produces a fine duplex microstructure and greatly increases corrosion resistance	Chong *et al.*[469]	1995
AISI 316L (Si_3N_4)	Laser surface alloying of AISI 316L stainless steel with silicon nitride improves the hardness to 300–1200 VHN and varied with Si content	Tsai *et al.*[470]	1994

Table 4 Continued

a. Laser surface alloying

System	Results	Reference	Year
Mild steel (stainless steel)	Stainless steel powder was coated on mild steel substrates using a plasma coating technique and laser melting. The oxidation rate in air (at 1073 K) decreased after plasma coating and was further improved by laser melting	Khanna *et al.*[471]	1992
Roping wires of grade BS 2763 and 1770	Laser surface alloying proved durable and friction/wear resistant as compared with thin coatings by physical vapour deposition. The rope compositions were about 0·65–0·75%C, 0·65–0·85%Mn and 0·15–0·35%Si. The coating material was Cr and Zr	Batchelor *et al.*[472]	1992
Low C steel	Laser surface alloying of low carbon steel with chromium resulted in an increase in surface hardness and resistance to abrasive wear compared with the bulk material	Tan and Doong[473]	1991

b. Laser surface cladding

System	Results	Reference	Year
Plain carbon steel	Laser surface cladding with 25%Cr+10%Ni formed a refined microstructure with a minimum dilution at the interface	Barnes *et al.*[474]	2003
Mild steel	A process for producing a dense, hard, iron–aluminide coating on a mild steel substrate has been successfully developed using pulsed laser assisted powder deposition	Corbin *et al.*[475]	2003
Carbon steel	In the present study, Ni–Cr–B–Si cladding on carbon steel has been achieved by high power diode laser. The microstructure is composed of a matrix of α-Ni with the presence of nickel (Ni_3Fe and Ni_3Si,), boron (B_4C and CrB) and chromium (Cr_3Si) intermetallics	Conde *et al.*[476]	2002
AISI 304 stainless steel	A comparison between plasma transferred arc welded and laser hardfaced Co based superalloy on AISI 304 stainless steel showed that, for normal temperature application, laser hardfacing shows the bests results; however, for high temperature application, plasma transferred arc deposits exhibited superior microstructure	d'Oliveira *et al.*[477]	2002

Liu *et al.*[516] laser surface melted Al–Si–graphite particulate composite. The wear resistance was improved because of microstructural refinement and improved microhardness. Watkins *et al.*[503] melted the surface of a series of Al transition metal alloys (Al–Cu, Al–Si, Al–Zn and Al–Fe) and obtained unique microstructural and compositional characteristics including improved hardness, wear and pitting corrosion resistance using a selected set of laser parameters. SiC dispersed Al alloy composite is the most promising class of metal matrix composite for applications in automobile and aircraft industries. However, these composites undergo severe degradation in chloride containing environment.

Sicard *et al.*[494] reported enhancement of mechanical and chemical properties of aluminium alloys (AlSi7Mg0·3) by excimer laser assisted nitriding treatment. The special nitriding treatment requires an excimer laser to irradiate the sample placed inside the cell with 1 bar nitrogen pressure allowing expansion of the vapour plasma, dissociation/ionisation of nitrogen by laser generated shock wave and subsequent penetration (adsorption and diffusion) of nitrogen from plasma up to some depth. While adequate laser fluence is required to create the plasma, this fluence must be limited below a threshold to prevent laser induced surface roughness. The resultant polycrystalline nitride layer is several micrometres thick and comprises pure AlN columns (200–500 nm thick) on top of equiaxed AlN grains in the diffusion layer. Heat conduction calculations show that a 308 nm laser is better suited for greater nitride thickness, as it corresponds to a weaker reflectance value for aluminium. Laser induced plasma formation was utilised to develop AlN coating on Al surface using XeCl laser (wavelength=308 nm) operated at 20 Hz repetition rate.[497] The vacuum chamber was filled with N_2 gas to a pressure of 760 torr. The interaction between the laser beam with metal surface led to melting of the substrate and reaction with the confined plasma causing the formation of AlN on the surface of Al. Wear resistance was improved by surface nitriding with a significant reduction in friction coefficient. However, it was observed that marginal presence of oxygen and greater degree of overlapping are beneficial for further improvement in wear resistance.

Laser surface alloying of 2014-T6 [Al–4·65Cu–0·5Mg–0·8Si (wt-%)] Al alloy with 25Cr–75Al, 75Cr–25Al, 25W–75Al, 75W–25Al, 8Al–72Zr–20Ni and 35Al–50Ti–15Ni powder mixtures was carried out using an Electrox 2 kW CW CO_2 laser.[503] The effects of alloying and overlapping on the microstructure and corrosion behaviour were studied. Similar attempt of laser surface alloying of Al with Mo (by injecting Mo powder into melt pool) was undertaken by laser surface alloying with a composition ranging from 14·8 to 19·1 wt-%Mo and resulted in 85–100 VHN in hardness and 84–92 GPa in Young's modulus (determined by nanoindentation).[502] Tang *et al.*[518] observed that laser surface alloying of manganese–nickel–aluminium bronze with Al was more effective in enhancing corrosion and cavitation erosion resistance in a 3·56 wt-%NaCl solution than that after laser surface melting. By employing polarisation resistance measurements under quiescent and cavitating conditions, the contributions

Table 5 Summary of selected studies on laser surface engineering of non-ferrous materials

a. Aluminium and its alloys

System	Results	Reference	Year
7075-T651 Al alloy	Laser surface melting with excimer (KrF) laser caused reduction in number and size of constituent particles and refinement of grain structure with an improved corrosion property	Chan *et al.*[493]	2003
AlSi7Mg0·3	The excimer laser nitriding process was developed to enhance the mechanical and chemical properties of aluminium alloys. The polycrystalline nitride layer composed of a pure AlN (200–500 nm thick) on an AlN (grains) in alloy diffusion layer was developed	Sicard *et al.*[494]	2001
Al–Li–Cu alloy (2·5Li, 1Cu and 0·12Zr)	The laser treated (using Nd:YAG laser) specimen exhibited improved oxidation resistance attributed to the compact oxide structure produced on its surface during oxidation	Hari Prasad and Balasubramaniam[495]	1997
Al–4%Cu	Residual stress changes from tensile to compressive due to laser surface treatment (using excimer laser). The microhardness is increased at a depth <10 mm from the surface	Badekas *et al.*[496]	1988
Al (AlN)	AlN coating was developed on Al substrate (using excimer laser) by laser induced plasma surface engineering with a significant improvement in wear resistance	Meneau *et al.*[497]	1998
6061 Al (TiC)	The four point bend test of Nd:YAG laser treated sample showed propagation of cracks from the surface and delamination of coating	Katipelli *et al.*[498]	2000
Al–Si (Al_2O_3–TiO_2)	A graded coating consisting of an Ni clad Al bond layer, a 50 wt-%Ni clad Al+50 wt-%(Al_2O_3–13 wt-%TiO_2) intermediate layer and an Al_2O_3–13 wt-%TiO_2 overlayer (or ceramic layer), was produced on an Al–Si substrate by plasma spraying and then was remelted by laser irradiation	Wang *et al.*[499]	1993
Al_3Ti or intermetallic compound matrix composite	Intermetallic compound Al_3Ti or intermetallic compound matrix composite (IMC) layers were formed on Al surface by CW CO_2 laser treatment. Intermetallic compound matrix composite surface layer improved the wear property of Al substrate	Uenishi and Kobayashi[500]	1999
Al (TiC)	TiC was deposited on 2024 and 6061 Al using Nd:YAG laser. The microstructure was found to vary with laser parameters	Kadolkar and Dahotre[501]	2003
Al (Cr)	Microstructure of laser surface alloyed Al with Cr contains relatively large amounts of intermetallic compounds dispersed in a matrix of α-A1	Almeida *et al.*[502]	1995
Laser surface alloying and laser surface melting of Al	A review of literature on laser surface melting and alloying of Al is summarised	Watkins *et al.*[503]	1998
Al7175-7351 (Cr)	Laser surface alloying of Al with Cr improved crevice corrosion resistance	Ferriera *et al.*[504]	1996
Al6013–SiC	Laser surface treatment of Al6013–SiC composite using a CW Nd:YAG laser resulted in selective vapourisation of the aluminium matrix with improved pitting resistance	Yue *et al.*[505]	1997
Al (Cr)	Vacuum deposited chromium films were irradiated in air using a Nd:glass laser. For a thin Cr deposition, Cr was evaporated; however, for a thick coating, it was uniformly distributed	Jain *et al.*[506]	1981
Al–4·3Cu	Al–4·3%Cu alloy was developed by laser surface alloying of Cr on Al substrate and subsequent remelting. The microstructure was investigated in details	Almeida *et al.*[507]	2001
Al–Cu	Mechanical and microstructural properties were found to vary with laser scan speed	van Otterloo and de Hosson[508]	1994
2014 Al	This corrosion behaviour of the overlapped areas in the laser melting and alloyed (with Cr, W, Zr–Ni or Ti–Ni) alloy was undertaken. Electrochemical testing showed microsegregation within the overlapped areas leading to the initiation of pitting corrosion	Watkins *et al.*[503]	1998
Al (Cu)	The microstructure and hardness variations throughout samples of an aluminum–copper alloy (Al–15 wt-%Cu) were analysed	Pinto *et al.*[509]	2003
Al–Fe alloy	Laser melting of Al–Fe alloy containing 0·25–8·0 wt-%Fe was carried out to study the effect of solidification rate on the microstructure	Gremaud *et al.*[510]	1990
Al–Si	Microstructure of the laser surface melted Al–Si alloy using a CO_2 laser was refined with extension of solid solubility of Si in Al. The corrosion resistance after laser treatment was improved in 10%H_2SO_4, and 10%HNO_3 solutions. However, laser treatment did not produce significant improvement in studies with 10%HCl and 5%NaCl solutions	Wong *et al.*[511]	1997

Table 5 Continued

a. Aluminium and its alloys

System	Results	Reference	Year
SiC reinforced Al AA 2009 (Al–3·8Cu–1·3Mg)	The corrosion resistance was significantly improved by laser surface melting attributed to r edistribution of $CuAl_2$ and extension in solid solubility of Cu in Al	Yue *et al.*[512]	1999
2024-T351 aluminium alloy	The improved corrosion resistance due to laser surface melting (using a CO_2 laser) is attributed to changes in the distribution and composition of the second phase particles present in the alloy	Li *et al.*[513]	1996
7075 Al alloy	Laser shock processing of A356, AlI2Si and 7075 aluminium alloys (using excimer laser) improved fatigue performance attributed to compressive residual stress levels	Peyre *et al.*[408]	2000
Al (SiC)	Al/SiC metal metrix composit layers were prepared by laser injection of SiC particles into Al substrate. An interface reaction layer consisted of Al_4C_3 plates forming a good mechanical bonding between the SiC particles and Al matrix	Vreeling *et al.*[514]	2000
Al	Al was irradiated in nitrogen atmosphere in pulsed nanosecond excimer laser and there was formation of AlN phase. The effect number of pulses and beam fluence on nitrogen incorporation was studied	Carpene and Schaaf[515]	2002
Al–Si–C composite	Laser surface melting of Al–Si–C composite was found to refine the grains with a significant improvement in hardness and wear resistance	Liu *et al.*[516]	1997
Al-6061	ZrO_2–7 wt-%Y_2O_3 powder mixture was plasma sprayed on Al followed by laser surface melting with an Nd:YAG laser. The wear resistance improved significantly	Fu *et al.*[517]	1997
Al (TiB_2)	Uniform *in situ* dispersion of TiB_2 was obtained by laser melting of a mixture of K_2TiF_6 and KBF_6 (2 : 1 weight ratio) on Al substrate, which improved hardness and wear resistance	Dutta Majumdar *et al.*[519]	2006

b. Copper and its alloys

System	Results	Reference	Year
Cu (Al)	Laser surface alloying of Al on Cu substrate using a ruby laser formed Al_2Cu phase. A theoretical model predicted cooling rate	Manna *et al.*[522]	1994
Zn coated Cu	Zinc coated copper specimens were irradiated with a CW CO_2 laser created surface alloys between zinc and copper on the surface of copper substrate; the morphology and its distribution were found to vary with laser intensity	Michaelides and Panagopoulos[523]	1993
Al bronzes (−5 at-%Fe, 13–28 at-%Al)	Laser melting of commercial Al bronze was conducted. A detailed microstructural and phase analysis was conducted	Vandenberg and Draper[524]	1984
Cu (Cr)	Laser surface alloying enhances resistance to adhesive/abrasive wear and erosion due to solid solution and dispersion hardening	Dutta Majumdar and Manna[525]	1999

c. Titanium and its alloys

System	Results	Reference	Year
Ti–6Al–4V (TiN)	TiN forms by laser surface alloying and improves wear resistance that occurs in three stages	Yilbas *et al.*[534]	2001
Ti (Si, Al and Si+Al)	Laser surface alloying with Si is more effective than Al/Si+Al to improve wear and corrosion resistance	Dutta Majumdar *et al.*[528–530]	2000 2002 1999
Ti–6Al–4V (WC and TiC)	Laser surface alloying with TiC improves, but that with WC deteriorates corrosion resistance	Cooper *et al.*[535]	1998
Ti (Ti+TiC)	Laser surface alloying with TiC significantly enhances the hardness of substrate Ti	Fouilland-Paille *et al.*[536]	1997
Ti and Ti–IMI829 alloy (N_2)	Laser surface alloying enhances hardness and corrosion resistance with a smooth surface finish	Weerasinghe *et al.*[537]	1996
Ti alloy (WC and TiC)	Laser surface alloying significantly enhances hardness	Masse and Mathieu[538]	1996

Table 5 Continued

d. Magnesium and its alloys

System	Results	Reference	Year
Stainless steel on Mg–17 vol.-%SiC composite	Laser cladding of stainless steel on ZK60/SiC composite (using a CW Nd:YAG laser) was achieved by thermal spray and subsequent laser remelting. The corrosion resistance was improved	Yue *et al.*[544]	2001
Al–12·5 wt-%Si	Laser cladding of Al–12·5 wt-%Si alloy onto a Mg/SiC composite (using a Nd:YAG laser) led to a significant improvement in corrosion resistance	Wang and Yue[545]	2001
17 vol.-%SiC/ZK60	Excimer laser surface treatment of Mg metal matrix composite caused dissociation of SiC	Wang and Yue[548]	2001
17 vol.-%SiC/ZK60	Excimer laser surface treatment of Mg–SiC metal matrix composite improved the corrosion resistance property significantly attributed to grain refinement. The improvement is more in nitrogen atmosphere	Yue *et al.*[541,544]	1997 2001
MEZ	Laser surface melting improves microhardness and corrosion resistance significantly by refinement of grains and supersaturation of rare earth element	Dutta Majumdar *et al.*[543]	2003
AZ91D and AM60B Mg alloys	Uniform and refined microstructure improves corrosion resistance	Dube *et al.*[539]	2001
AZ91 Mg alloy	Pulsed excimer laser irradiation improves corrosion resistance	Schippman *et al.*[540]	1999
Mg–ZK60+SiC composite	Uniform and refined microstructure produced by excimer laser improves corrosion resistance	Yue *et al.*[541]	1997

of erosion induced corrosion and corrosion induced erosion were determined.

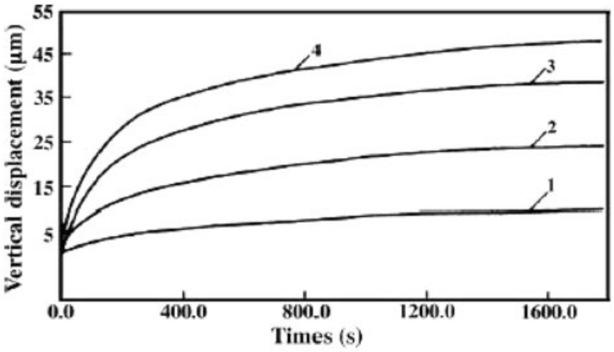
Jain *et al.*[506] vacuum deposited Cr film (thickness varying from 650 to 1350 mm) on Al substrate and irradiated with single pulses (with a full width at half maximum of 7 ns) from a Nd:glass laser operating in a TEM00 mode at peak power densities of 6000–15 000 GW m^{-2} in air. Rutherford backscattering of 3·05 and 4·8 MeV He^+ ions was employed to determine depth profiles of chromium in aluminium. Almost complete removal of chromium at 8000 J m^{-2} is followed by the rapid diffusion of chromium (in molten Al) with an accompanying reduction in evaporation loss at higher energy densities. Almeida *et al.*[507] carried out laser surface alloying of Al with Cr using powder injection method with a laser beam power of 2000 W and a scan speed of 5 mm s^{-1}. Cr content in the alloyed zone was 4·3 wt-%. The microstructure of the alloyed zone consists of dispersion of long faceted intermetallics of Al–Cr organised radially along primary α-solid solution matrix. The microhardness of the alloyed layer is significantly improved and increased with Cr content in the alloyed zone. Similarly, Ferreira *et al.*[504] also reported the improvement in crevice corrosion in Al 7175-T7351 alloy by laser surface alloying with Cr.

Dutta Majumdar *et al.*[519] developed a titanium boride dispersed Al based metal matrix composite on the surface of pure Al by melting the surface of as received substrate using a CW CO_2 laser and simultaneous deposition of a mixture of K_2TiF_6 and KBF_6 (in the weight ratio of 2 : 1) through an external feeder (at a feedrate of 4 g min^{-1}). The microstructure of the surface layer consisted of uniformly dispersed titanium boride (TiB) and titanium diboride (TiB_2) particles in grain refined Al matrix. Figure 15 shows the variation in wear loss (in terms of depth of wear) as a function of time for as received and laser composite surfaced Al with TiB_2 (lased with a power of 1·2 kW and a scan speed of 700 mm min^{-1}) using a friction and wear monitor unit with applied loads of 500 and 900 g respectively. It is evident that the rate of wear increases both with time and applied load. In pure Al (curves 3 and 4), the wear rate is very high in the initial stage (up to 5 min) following which the same decreases.

On the other hand, the magnitude and rate of wear are significantly lower in laser composite surfaced Al than those of the as received Al, although the same increases with increasing load for both as received and laser composite surfaced Al with TiB_2. The kinetics of wear was, however, found to vary with laser parameters. Maximum wear resistance was achieved in laser composite surfaced Al lased at a power of 1·2 kW and a scan speed of 700 mm min^{-1}. The improved wear resistance of laser composite surfaced Al was attributed to improved microhardness of the composite layer because of both grain refinement and precipitation of fine and hard TiB_2 and TiB particles in the matrix.

An aluminium oxide layer of 100 μm in thickness has been successfully coated on aluminium alloy 6061 and pure aluminium using a powder mixture of silicon oxide and aluminium by laser cladding using a CW CO_2 laser. A strong Al/Al_2O_3 interface was formed possibly due to an exothermic chemical reaction between SiO_2 and Al.[520] Laser cladding on pure Al substrate was carried out using a CW Nd:YAG laser and a coaxial powder injection system to develop composite coatings of 0–40 wt-%Si and 0–30 wt-%TiC particles in Al matrix.[521]

The microstructure of the coatings was homogeneous and crack/pore free with uniform distribution of carbides. The addition of Si and TiC reinforcement particles increased the hardness and wear resistance of the coatings. Katipelli *et al.*[498] carried out a four point bend test on laser surface engineered TiC coating on 6061 Al alloy to study the interfacial strength or integrity of the coating including strain energy release rate and stress intensity factor. It was inferred that the



15 Cumulative wear loss (in terms of vertical displacement) as function of time for as received Al with applied loads of 500 g (plot 3) and 900 g (plot 4) and laser composite surfaced Al with TiB_2 (with power of 1·2 kW and scan speed of 700 mm min^{-1}) with applied loads of 500 g (plot 1) and 900 g (plot 2) using friction and wear monitor unit[519]

interface plays a key role in determining the mechanical behaviour of ceramic coating system. Similar investigations on the laser surface alloying of TiC on aluminium (2024 and 6061) alloys by Nd:YAG laser showed the effect of process parameters on the microstructures of coating.[501]. Fu *et al.*[517] studied the effect of laser surface melting on the wear behaviour of plasma sprayed ZrO_2 coating on Al. Porosity and roughness of the plasma sprayed coating were significantly reduced and the bond strength between the coating and substrate was increased by laser melting. However, the presence of cracks was observed despite the fact that wear resistance was significantly improved after laser remelting.

Wang *et al.*[499] developed a graded coating consisting of a Ni clad Al bond layer, a 50 wt-%Ni clad Al+50 wt-% (Al_2O_3–13 wt-%TiO_2) intermediate layer and an Al_2O_3–13 wt-%TiO_2 overlayer (or ceramic layer) on an Al–Si alloy substrate by plasma spraying followed by laser surface remelting. While the plasma sprayed coatings seem prone to spallation at the different depths originating from macrocracks in the ceramic layer during thermal cycling, the laser remelted coatings, containing a network of microcracks in the ceramic layer reveal a significantly reduced tendency of spallation at the interface between the intermediate and bond layer. Thus, laser remelting improved the spalling resistance of plasma sprayed coating to thermal shock, although cracking due to thermal shock remains a problem to be solved. Uenishi and Kobayashi[500] obtained Al_3Ti dispersed intermetallic matrix composite on Al by laser surface cladding.

Laser surface alloying of copper alloys

Laser surface alloying of copper and its alloys has been extensively studied by several researchers and are summarised in Table 5b. Manna *et al.*[522] made an attempt of laser surface alloying of Cu with Al (predeposited by physical vapour deposition, and subsequently laser melted) using ruby laser (irradiated in air for varying numbers of 30 ns pulses with 0·2 J mm^{-2} energy density) and identified the phases through careful XRD and photoelectron spectroscopy. Michaelides and Panagopoulos[523] irradiated Zn coated Cu with a CW CO_2 laser to produce a thin Zn–Cu surface alloy on copper substrate. The amount and distribution of zinc in the alloyed zone and the intensity of the plasma created on the surface during laser irradiation were found to be a function of the laser power density and interaction time. Vanderberg and Draper[524] laser surface melted commercial Al bronzes (containing 5 at-%Fe and 13–28 at-%Al) and found a variety of phases formed in near surface region. The microstructure consisted of α-phase (face centred cubic solid solution), β_1 phase (product of ordering transformation of the high temperature disordered body centred cubic phase β) and even β phase (martensite formed by shear transformation on quenching). Rapid quenching in laser treatment also resulted in β_1 retention (by suppressing of martensite change) at Al concentration that was lower than what was usually observed with splat quenching experiments.

Dutta Majumdar and Manna[525] attempted to enhance the wear and erosion resistance of Cu by laser surface alloying with Cr (electrodeposited with 10 and 20 μm thickness t_z). Total Cr content X_{Cr}, Cr in the form of precipitates f_{Cr} and Cr in solid solution with Cu Cu_{Cr} in the alloyed zone were determined by energy dispersive spectrometry, optical microscope and XRD technique respectively. Laser surface alloying extended the solid solubility of Cr in Cu as high as 4·5 at-% as compared with the equilibrium solid solubility limit of 1 at-%.

The microhardness of the alloyed zone was found to improve significantly (as high as 225 VHN) following laser surface alloying as compared with 85 VHN of the base metal.[525] Since hardness is related to Cr present in solid solution and dispersed as precipitates in the matrix, the variation in average microhardness of the alloyed zone as a function of the total Cr content, Cr dissolved in solid solution or volume fraction of Cr precipitated shows that hardness increases with all these microstructural factors, especially with the degree of solid solubility extension of Cr in Cu.[525] A significant enhancement on wear resistance and erosion resistance (both at room temperature and at elevated temperature) properties was also achieved by laser surface alloying.[525] Zhao *et al.*[526] studied the effect of laser surface treatment on the surface energy of copper plate in terms of the surface microstructure analysis and theoretical computation. The surfaces of the copper plates were treated by Nd:YAG pulsed laser with different powers. The microstructures of the treated copper plates were analysed by optical microscopy and XRD, and the wetting experiment was performed to evaluate the variation in surface energy. The surface energy of the treated surface was found to vary with the applied laser power.

Laser surface alloying of titanium and its alloys

Tian *et al.*[527] laser surface alloyed pure Ti with TiN–B–Si–Ni powders and detected the formation of different kinds on intermetallics in the alloyed zone with a very high microhardness (1500–1600 HV0·1), a low friction coefficient ($\sim$0·4) and a higher oxidation resistance than the untreated substrate.

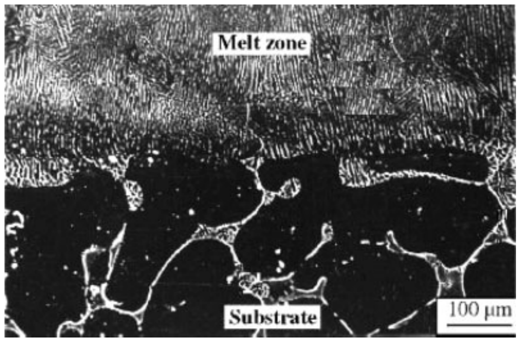
Laser surface alloying of Ti with Si, Al and Si+Al (with ratios of 3 : 1 and 1 : 3 respectively) was conducted to improve the wear and high temperature oxidation resistance of Ti.[528–530] Isothermal oxidation studies conducted at 873–1023 K showed that laser surface alloyed Ti with Si and Si+Al significantly improved the isothermal oxidation resistance. The effect of laser surface alloying of Ti with Si or Si+Al on wear resistance was also studied. Under comparable conditions of scratching, Ti undergoes the most rapid wear loss followed by that in laser surface alloyed specimens. The laser alloyed sample with Si undergoes the minimum wear loss. The improved wear resistance of laser surface alloyed Ti with Si was attributed to the formation of hard Ti_5Si_3 precipitates in the alloyed zone.[528] Manna *et al.*[531,532] made an attempt to develop an alternative electrode material for neural stimulation electrode by laser surface alloying of Ti with Ir that can mimic the normal spatiotemporal pattern of neuronal activation by reversible charge transfer. The usual electrode made of iridium is expensive, brittle and not amenable to miniaturisation by plastic deformation. In comparison, titanium is cheaper, biocompatible and amenable to drawing/etching. A few relevant literatures on the laser surface alloying of titanium and its alloys are summarised in Table 5c.[528–538]

Laser surface alloying of magnesium and its alloys

Magnesium and its alloys are widely used in automotive and aerospace applications because of its lightweight. However, poor wear and corrosion properties are of serious concern for its application as structural components. Laser surface melting of magnesium and its alloys was reported to be effective to refine the microstructure and improve corrosion resistance in AZ91D/AM60B,[539] AZ91,[540] Mg-ZK60[541] and other commercial Mg based alloys.[542] Detailed studies on kinetics and mechanism of laser surface melted MEZ (a Mg–Zn alloy) were reported by Dutta Majumdar *et al.*[543] Figure 16 shows the cross-section of the laser surface melted MEZ alloy, lased with a power of 2 kW and a scan speed of 200 mm min^{-1}, consisting of fine columnar grains growing epitaxially from the liquid/solid interface. The underlying substrate grains are significantly coarser with grain boundaries decorated with thick films of Mg/Zr/Ce rich compound (confirmed by energy dispersive spectroscopy). Furthermore, the melted zone/substrate interface is crack/defect free and well compatible with practically no noticeable amount of heat affected zone.

A detailed energy dispersive spectroscopy analysis of the melt zone suggests that laser melted zone has a more homogeneous elemental distribution than that in as received MEZ. Figure 17 presents the microhardness profiles of the laser surface melted MEZ specimens as a function of vertical distance from the surface measured on the cross-sectional plane and shows that the microhardness of the melted zone has significantly increased by 2–4 times (85–100 VHN) than that of the substrate (35 VHN) primarily due to grain refinement and solid solution hardening. While average microhardness varies with the concerned laser parameters, the maximum microhardness is achieved following laser surface melting with a power of 1·5 kW and a scan speed of 200 mm min^{-1}.

Figure 18 shows the kinetics of pit formation in the as received and laser surface melted (with a power of 2 kW and a scan speed of 200 mm min^{-1}) MEZ samples subjected to standard immersion tests in a 3·56 wt-% NaCl solution. It is evident that both the extent and rate of pitting are significantly reduced following laser surface melting. Furthermore, the rate of pitting is initially faster and decreases after 30 h of immersion in as received MEZ, while visible pits are observed in laser surface melted MEZ only after 12 h of exposure revealing a much slower kinetics of the process. In fact, the pits in the laser surface melted samples are both

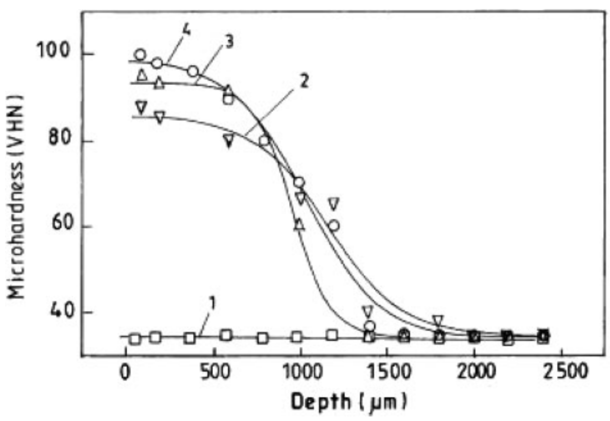


16 Scanning electron micrograph of cross-section of laser surface melted MEZ sample (lased with power of 2 kW and scan speed of 200 mm min^{-1}): defect free interface with significant grain refinement in laser melted zone may be noted

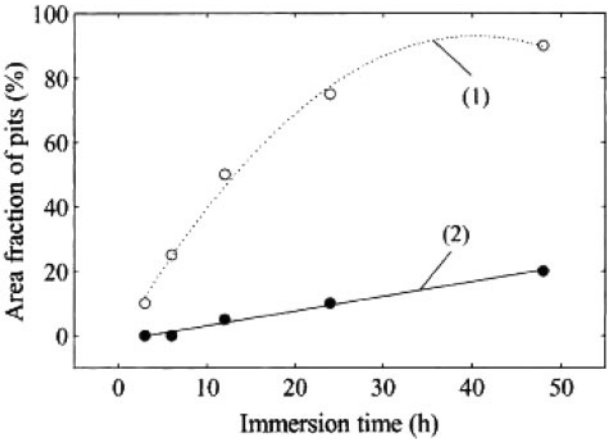
smaller in dimension and remain isolated compared with larger and interconnected pits in as received MEZ. Yue *et al.*[544] surface melted Mg–SiC (17 vol.-%) composite using KrF excimer laser with Ar and N_2 as shrouding environment and achieved considerable improvement in corrosion resistance in 3·56%NaCl solution primarily due to microstructural refinement. Similar surface melting in N_2 atmosphere improved the corrosion resistance further because of the formation of magnesium nitrides.

Laser surface alloying was also attempted on magnesium and its alloys to improve the wear and corrosion resistance properties. Wang and Yue[545] have achieved significant improvement in corrosion resistance of SiC dispersed Mg by LSC with Al–12Si alloy layer.

Ignat *et al.*[546] studied the effect of laser surface alloying on surface alloying of WE43 and ZE41 Mg alloys with Al using a 3 kW CW Nd:YAG laser and found that the alloyed zone microstructure consisted of Al_3Mg_2 and $Al_{12}Mg_{17}$ intermetallics with an improved microhardness and corrosion property. Similar investigations of laser surface alloying of MEZ with Al+Mn (in 3 : 1 and 1 : 1 by weight) using a CW CO_2 laser significantly improved the wear and corrosion properties of the base alloy.[547] Several other works related to the surface engineering of magnesium and its alloys are summarised in Table 5d.[539–548]



17 Microhardness profile as function of depth from surface of (1) as received MEZ and laser surface



18 Kinetics of pit formation (in terms of area fraction of pits as function of immersion time) in (1) as received and (2) laser surface melted (with power of 2 kW and scan speed of 200 mm min^{-1}) MEZ samples followed by immersion test in 3·56 wt-%NaCl solution[543]

Laser assisted CVD

Chemical vapour deposition encompasses a group of processes that involve depositing a solid material from a gaseous phase. Laser chemical vapour deposition (LCVD) involves techniques in generating solid deposits on the surface of a substrate by dissociation of reactants followed by heating using laser beam. Laser chemical vapour deposition deposits materials at a faster rate with a high accuracy, a low porosity and a high degree of crystallinity and hence, a coating with an excellent mechanical properties and thermal stability. Pyrolytic LCVD is a thermally driven process in which the energy of a laser beam is used to heat the surface of a substrate to the temperature required for chemical deposition. Pyrolytic LCVD has been applied to deposit a large varieties of materials including Cu, Au, Si, Al, W, C, WC, B, TiC, TiN, SiC and Si_3N_4.[549–552] However, a careful control of process parameters is extremely important when depositing multiple layers of materials.[553] Since the underlying deposit may exhibit irregularities, proper control is needed to insure that these irregularities are not amplified with each successive layer. Noel and Kovar deposited titanium carbide (TiC) coatings onto tantalum substrates using hydrogen gas, titanium tetrachloride ($TiCl_4$) and either methane (CH_4) or acetylene (C_2H_2) mixtures.[554,555] The kinetics of deposition was found to vary with precursor composition. It was also observed that the phases (both quantity and area fraction) and orientation of the deposition varied with precursor composition and temperature. Laser chemical vapour deposition was also reported to develop oxide based ceramic coatings such as partially yttria stabilised zirconia and titania (TiO_2).[556] The deposition rate was significantly higher than that of the conventional chemical vapour deposition which is attributed to the plasma formation around the deposition zone, above critical values of laser power and substrate preheating temperature. The activation energy for the deposition for both yttria stabilised zirconia and TiO_2 films was $\sim$10 kJ mol^{-1}, suggesting a mass transfer limited process. A wide variety of film morphologies were obtained based on the applied parameters. The columnar structure contained a large amount of nanopores at columnar boundary and inside grains. A detailed thermodynamic calculation was undertaken to understand the process sequences. On the other hand, a kinetic model is essential to determine the apparent activation energy and the apparent order of reaction.[556]

Laser surface engineering of ceramic materials

Laser surface engineering has also been extensively studied to tailor the surface of ceramic materials.[557–569] Heras *et al.*[557] attempted laser assisted removal of water repellent substances or coatings on stone and similar materials used in buildings or sculptures of artistic value. The effect of wavelength on laser assisted removal of two water repellent coatings or treatments applied on limestone, namely, Paraloid B-72, a copolymer of methyl acrylate and ethyl methacrylate, and Tegosivin HL-100, a modified polysiloxane resin, was investigated by using the four harmonics of a Q switched Nd:YAG laser (1064, 532, 355 and 266 nm). It was observed that the removal of the thin coatings depended on the laser irradiation conditions and on the characteristics of the coatings. The fundamental laser radiation was effective in removing both the coatings but thermal alteration processes were induced on the constituent calcite crystals. The best results were obtained by irradiation at the near UV wavelength of 355 nm.

Lawrence and Li[558] glazed the surface of Al_2O_3 based refractory by means of high power diode laser and augmented the wear rate and wear life characteristics within both normal and corrosive (NaOH and HNO_3) environmental conditions. Life assessment testing revealed that diode laser generated glaze increased the wear life of the Al_2O_3 based refractory by 1·2–13·4 times depending upon the environmental conditions. Such improvements are attributed to the fact that the microstructure of the Al_2O_3 based refractory was altered from a porous, randomly ordered structure to a much more dense and consolidated structure after laser treatment that contained fewer cracks and porosities.

Sciti *et al.*[561] surface treated alumina and silicon carbide ceramics with a KrF excimer laser and studied the influence of laser fluence (1·8 and 7·5 J cm^{-2}), number of laser pulses (1–500), frequency (1–120 Hz), pulse duration (25 ns) and shrouding environment on the characteristics of the treated surface. Microstructural analyses of the surface and cross-section of the laser processed samples at a low fluence (1·8 J cm^{-2}) evidenced that the surface of both ceramics is covered by a scale due to melting/resolidification. At a high fluence (7·5 J cm^{-2}), there are no continuous scales on the surfaces as material is removed by decomposition/vapourisation. On the alumina surface, a network of microcracks formed, while on the silicon carbide, products of different morphologies (flat and rugged areas, deposits of debris and discontinuous thin remelted scales) were detected.

Schmidt and Li[563] studied the influence of laser glazing of sintered alumina as well as alumina–silica mixtures on the surface characteristics and service life using a high power diode laser and a high speed motion analysis system for use in a multitude of applications as insulators, crucibles, vacuum system windows, etc. The laser glazed material seems robust and exhibits good adherence to the bulk ceramic.

Stoltz and Poprawe[564] modified the surface conductivity of dielectric ceramics by CW and pulsed laser irradiation. Conductive structures on Al_2O_3 were developed with a CO_2 laser and on AlN with an excimer laser irradiation (pulse width=25 ns). Laser treated specimens were subjected to a detailed characterisation of electrical resistivity, chemical composition, structure and morphology.

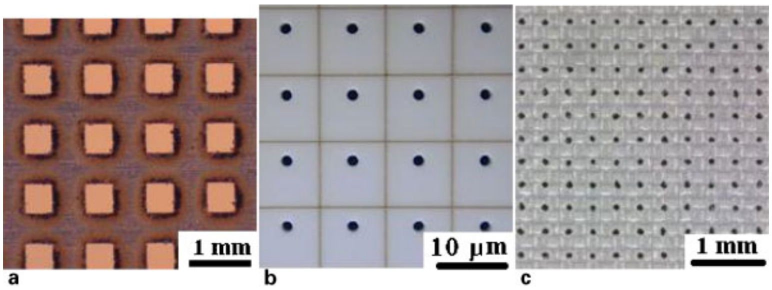
Polycrystalline ceramics, notably alumina (polycrystalline aluminium oxide) are used widely in the electronics industry as substrates for metal film deposition, as well as in other industrial applications. Multishot pulsed XeCl excimer laser irradiation of commercial fine grained polycrystalline alumina substrates is found to significantly improve the bonding characteristics between the metal and film.[567] Laser irradiation produces a smoother surface finish increasing the adhesion strength of subsequently deposited copper films to the laser treated alumina surface by a factor of 3–5 under optimum lasing conditions. X-ray photoelectron spectroscopy measurements suggest that the electrical activation of the near surface region may also contribute to the enhanced copper adhesion.

Buerhof *et al.*[568] studied the effect of laser ablation on surface modification of glass using a 50 W CO_2 laser (10·6 pm) and a 40 W/2 J/20 Hz XeCl excimer laser (308 nm). The infrared radiation heated the sample surface locally while high power UV radiation caused ablation of glass with negligible thermal effects. Removal rates of various glasses during laser ablation have been determined.

Laser surface engineering of polymers

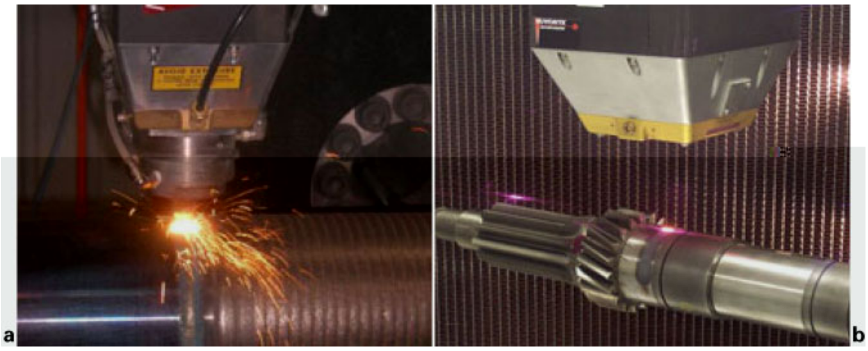
Laser induced surface modification and laser induced patterning of polymeric surfaces by deposition, etching and ablation have drawn considerable attention of late. Wong *et al.*[570] irradiated the surface of polyethylene terephthalate with KrF excimer laser (having a wavelength of 248 nm) and studied the surface image using atomic force microscopy. The surface roughness and ripple spacing increased with laser energy. Merging of the ripples (due to surface tension forces) is the driving force for the increased ripple spacing. With appropriate laser treatment, the hydrophobicity of polyester can be greatly enhanced, making the treated polyester highly unwettable. This provides the textile industry a new method for making a high tech water repellent textile product. Adhi *et al.*[571] irradiated the surface of poly-tetra-fluoro-ethylene films by femtosecond UV radiation from an excimer laser (KrF: λ: 248 nm; t_p: 380 fs) in air and investigated the surface using X-ray photoelectron spectroscopy, and bulk by Fourier transform infrared spectroscopy. There was no signature of incorporation of hydrogen and/or oxygen, or the formation of a cross-linked network of carbon indicating chemically clean processing in contrast to nanosecond excimer laser processing which chemically degrades the surface. Huang *et al.*[572] discussed a method of improving adhesion to fluorocarbon resin by irradiation with an XeCl excimer laser. The respective adhesive force between the polymer and metal, and contact angles with water were measured by the shear test method and a Ne–He laser system. The adhesive force improved when the polymer surface was irradiated in the presence of water or solutions of boric acid, sodium hydroxide, copper sulphate and sodium aluminate. These results were interpreted in terms of a simple laser heating model. Csete *et al.*[573] generated a new, two-dimensional grating-like structure on spin coated films made of polycarbonate which might be used as a special kind of mask.

Drinek *et al.*[574] photolysed hexamethylcyclotrisilazane with an ArF excimer laser at 84 K. At room temperature and in air, the film changed into a polymer having siloxane units. However, fragments generated by laser irradiation of this polymer in the frozen film made a similar polymer film having a siloxane structure on a polyvinyl alcohol film in air. The polymeric film with siloxane units prepared by the cryogenic laser ablation method displayed hydrophobic properties. Lu *et al.*[575] surface modified polyimide film by pulsed UV laser. Irradiation caused a photochemical reaction in the hydrophilic groups. In addition, a ripple microstructure was formed on the surface when the angle of incidence of the laser beam was 20–50°. Csete and Bor[576] generated laser induced periodic surface structures on oriented and amorphous thick, as well as on spin coated thin, polycarbonate films by a polarised ArF excimer laser. Studies on the influence of the film structure and thickness on the development of periodic surface structure revealed that the line shaped structure would transform into droplets below a critical thickness of the spin coated direction of the electric field are obtained with the ArF or KrF radiation when it is strongly absorbed by the polymer. Several polymers in thin or thick films of commercial or laboratory origin were textured in air or in vacuum or both depending on the chemical structure. Fluence should be below the ablation threshold and must be chosen in a narrow window, which depends on the polymer and the wavelength used. Surface characterisation by scanning and transmission electron microscopies and ellipsometry reveals a good insight



a 500 μm square holes with 500 μm gaps in between drilled into 200 μm thick copper foils by pulsed Nd:YAG laser; *b* 1 mm diameter holes with 10 μm wide lines scribed on alumina sheet by CO_2 laser; *c* high magnification view of array of 150 μm diameter holes drilled in fabric with a spacing of 625 μm by CO_2 laser

19 Examples of ultrafine drilling[586]



a laser cladding a shaft with Satellite 6; *b* laser transformation hardening a transmission shaft

20 Examples of CW CO_2 laser assisted[586] forming/manufacturing of large engineering component

manufacturing using robot controlled stage and pneumatic/mechanically driven delivery system (of wire, powder and sheet) has enabled the creation of large components practically of any desired dimension and geometry. The versatility is further extended in the ability of laser to simultaneously heat, add, join or remove materials of dissimilar nature such as conducting and non-conducting, hard and soft, and dense and porous. Laser is now utilised to close surface porosities on coated and welded products and also for creating graded aggregate with controlled porosity, composition and microstructure. This microstructural or compositional gradation can be achieved within very narrow to wide length scales and along horizontal or vertical directions.

In order the showcase some of the recent exploits of laser that are unique to laser material processing, we pick up three examples from commercial products and practices. Figure 19*a–c* demonstrates the unique capability of laser to produce ultrafine holes in square, rectangular or round shapes in metals, ceramic or polymeric materials. Besides drilling, the same material can be marked, scribed or machined using the same tool and facility. The precision and dimensional accuracy of these holes are unmatched notwithstanding the advantage of fully automated processing at an extremely fast speed and productivity. The holes have completely vertical walls with no distortion of the bulk workpiece. Holes with tapered inclination, threads or steps are also possible. Similar perforations can be made on thin filter or cigarette papers, round rims or flat turbine blades with equal ease and precision.

Laser drilling can be used to produce microholes with very high accuracy and close tolerance in almost all solids (metals, ceramics, diamond, silicon and other semiconductors, polymers, glass and sapphire) in the form of pinholes, nozzles, orifices, vias, photovoltaic cells, fuel injection nozzles, vertical probe cards, metered dose inhaler products, slits for scientific instrumentation, inkjet printer nozzles, detectors, sensors, high resolution circuitry, fuel cells, fibre optic interconnects and medical devices.

Figure 20*a* and *b* demonstrates how laser can be used for developing a clad or over layer on large shaft and heat treating (transformation hardening) transmission shaft, axle rods or gears. Similarly, laser cladding is exploited in refurbishing or reclaiming rolls, tools, blades, discs and turbine components. Transformation hardening can be applied mostly to steels and cast irons, but the ultrafine grain structure produced by surface melting and associated rapid cooling can also produce hard and wear resistant surfaces on non-ferrous metals and alloys. Controlled laser irradiation can produce localised heating, drilling vias and holes, annealing and even metallisation and doping in semiconductor wafers and films.

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